

# Joint Online Optimization of Power Allocation and Task Scheduling for Data Offloading in LEO Satellite Networks

Lijun He<sup>1</sup>, Member, IEEE, Zheyuan Li<sup>2</sup>, Juncheng Wang<sup>3</sup>, Member, IEEE, Ziye Jia<sup>4</sup>, Member, IEEE, Yanting Wang, Member, IEEE, Chau Yuen<sup>5</sup>, Fellow, IEEE, and Zhu Han<sup>6</sup>, Fellow, IEEE

**Abstract**—The rapid expansion of Low Earth Orbit (LEO) satellites inescapably leads to the explosive growth of space data in LEO Satellite Networks (LSNs). The stochastic nature of space data arrivals and the intrinsically time-varying satellite-ground links in LSNs pose significant challenges for offloading substantial volumes of space data from LSNs to the ground stations. To overcome these challenges, we systematically study the joint online optimization of power allocation and task scheduling for data offloading in LSNs. Firstly, we remove the constraint of mean rate queue stability from the formulated joint online optimization problem and leverage Lyapunov optimization to decouple it into a set of per-time-slot subproblems. Each subproblem is then divided into a task scheduling problem and a power allocation problem. Subsequently, we derive a closed-form optimal solution for the power allocation problem, and a multi-armed bandit-based quasi-optimal solution for the task scheduling problem. Furthermore, we extend the aforementioned solutions to address the original joint online optimization problem. Through theoretical analyses, we show that the proposed algorithms consistently attain a sublinear time-averaged regret. Extensive simulation results demonstrate that our proposed algorithms exhibit superior performance over other benchmarks.

**Index Terms**—LEO satellite networks, online data offloading, power allocation, task scheduling, multi-armed bandit.

## I. INTRODUCTION

THE exponential proliferation of wireless networking services continuously highlights the need for seamless global coverage and increased data transmission rates in prospective 6G communication networks [1]. Deploying costly base stations and other communication facilities to achieve this seamless global coverage with high-speed data transmission is impractical. Low-earth-orbit Satellite Networks (LSNs), owing to their extensive coverage and elevated communication rates, are progressively emerging as a feasible supplementary solution to traditional terrestrial networks, promising ubiquitous connectivity for 6G [2], [3], [4]. With the rapid decrease in satellite launch costs, tens of thousands of Low Earth Orbit (LEO) satellites are planned to be deployed in space through various commercial satellite projects, such as SpaceX's Starlink, Eutelsat Group's OneWeb, and Telesat's Lightspeed constellation [5], [6], [7]. Currently, LSNs provide extensive coverage services across a wide range of sectors, including rescue and disaster management, military operations, and environmental monitoring [8], [9], [10]. The accelerated proliferation of LEO satellites has led to an explosive growth of space data in LSNs [11]. A substantial portion of space data necessitates offloading from LEO satellites to Ground Stations (GSs) for subsequent processing and analysis [12]. However, the rapid movement of LEO satellites results in intermittent communication between network nodes in LSNs [13]. Thus, the key to improving the data offloading efficiency of LSNs lies in devising an effective data offloading scheme that can reconcile the burgeoning growth of space data with limited and intermittent communication resources.

The major challenges to designing an efficient data offloading scheme for LSNs are of the following three facets: 1) *Stochastic space data arrivals*. Space data typically arrives at LSNs in a random and unpredictable manner. This unpredictability is influenced by several factors, such as the application fields of satellites, stochastic events, and variations in data traffic types. Accommodating the stochastic arrivals of space data in real time poses a significant challenge for the efficient design of data offloading in LSNs. 2) *Time-varying Satellite-Ground Links (SGLs)*. The long communication distance between LEO

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Lijun He is with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China, and also with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: lijunhe@cumt.edu.cn).

Zheyuan Li and Yanting Wang are with the School of Software, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: schwarz@mail.nwpu.edu.cn; yantingwang@nwpu.edu.cn).

Juncheng Wang is with the Department of Computer Science, Hong Kong Baptist University, Hong Kong (e-mail: jcwang@hkbu.edu.hk).

Ziye Jia is with the College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China (e-mail: jiaziye@nuaa.edu.cn).

Chau Yuen is with the School of Electrical and Electronics Engineering, Nanyang Technological University, Singapore 639798 (e-mail: chau.yuen@ntu.edu.sg).

Zhu Han is with the Department of Electrical and Computer Engineering, University of Houston, Houston, TX 77004 USA, and also with the Department of Computer Science and Engineering, Kyung Hee University, Seoul 446-701, South Korea (e-mail: zhan2@uh.edu).

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satellites and GSs results in a large communication latency. Furthermore, the capacity of SGLs is significantly affected by atmospheric precipitation, including clouds, fog, and rain [14]. As a result, accurately acquiring real-time atmospheric precipitation information and predicting the current channel state of SGLs in LSNs is particularly challenging in the presence of high communication latency [15]. 3) *Trade-off between network metrics*. The large amount of offloaded space data inevitably leads to excessive energy consumption on LEO satellites. These satellites, however, are powered by energy-constrained batteries that necessitate long-term energy consumption control. Consequently, the challenge lies in balancing energy consumption with the volume of offloaded data over time in LSNs.

To address the above challenges, we jointly optimize power allocation and task scheduling for LSNs in an online manner, to maximize the total amount of offloaded data, while adhering to the constraint on time-averaged energy consumption. The main contributions of this paper are outlined below:

- We formulate an online optimization problem to enable joint optimization of power allocation and task scheduling in LSNs. Our goal is to maximize the long-term time-averaged volume of offloaded data while subjecting to the energy consumption constraint. Unlike prior works on online data offloading for LSNs, our joint online optimization problem efficiently accommodates both stochastic space data arrivals and time-varying SGLs under the long-term time-averaged energy consumption constraint.
- We provide an efficient solution for the formulated joint online problem for data offloading in LSNs. Firstly, we address a simplified version of our formulated problem, employing Lyapunov optimization to transform it into a series of per-time-slot subproblems. We then divide each subproblem into a task scheduling problem and a power allocation problem. By exploiting the special structures of the power allocation and task scheduling problems, we derive an optimal solution for power allocation and a Multi-Armed Bandit (MAB) based quasi-optimal solution for task scheduling. Furthermore, we extend the derived solutions to develop an efficient two-layer online solution for the original problem. Finally, theoretical analyzes demonstrate that the proposed algorithms produce a time-average regret of  $O(1/V + 1/T + \sqrt{\ln T/T})$ , where  $T$  represents the total number of time slots and  $V$  denotes an adjustable positive parameter.
- We conduct extensive experiments on the LSNs to demonstrate three main insights in the following. Firstly, the joint online optimization of power allocation and task scheduling can significantly increase the total amount of data offloaded from the LSNs. Secondly, our proposed algorithms efficiently balance the energy consumption and the amount of data offloaded in LSNs. Thirdly, our proposed solutions not only accommodate both stochastic space data arrivals and time-varying SGLs, but also significantly outperform existing state-of-the-art alternatives across diverse parameter configurations.

The remainder of this paper is organized as follows. In Section II, we review the related work on the data offloading

TABLE I  
NOTATION LIST

| Notation               | Description                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------|
| $\mathcal{U}$          | Set of LEO satellites                                                                         |
| $\mathcal{G}$          | Set of GSs                                                                                    |
| $\mathcal{T}$          | Set of time slots (i.e., $\mathcal{T} = \{1, \dots, T\}$ )                                    |
| $\tau$                 | Duration of one time slot                                                                     |
| $\mathcal{G}_u(t)$     | Subet of $\mathcal{G}$ that can be connected by LEO satellite $u$ in time slot $t$            |
| $a_u(t)$               | Data amount generated for the task on LEO satellite $u$ in time slot $t$                      |
| $R_{u,g}(t)$           | Achievable data rate of SGL $(u, g)$ in time slot $t$                                         |
| $R_u(t)$               | Offload transmission rate of LEO satellite $u$ in time slot $t$                               |
| $B_c$                  | Bandwidth                                                                                     |
| $\gamma_{u,g}(t)$      | Signal-to-noise ratio of SGL $(u, g)$ in time slot $t$                                        |
| $P_{u,g}(t)$           | Transmit power (in Watts) allocated to the SGL $(u, g)$ in time slot $t$                      |
| $\sigma_g^2(t)$        | Noise power at GS $g$ in time slot $t$                                                        |
| $\Psi_{u,g}(t)$        | Channel gain of SGL $(u, g)$ in time slot $t$                                                 |
| $h_u^t(t)$             | Transmit antenna gain of LEO satellite $u$ in time slot $t$                                   |
| $h_g^r(t)$             | Receiver antenna gain of GS $g$ in time slot $t$                                              |
| $\xi_{u,g}(t)$         | Path loss (in dB) between LEO satellite $u$ and GS $g$ in time slot $t$                       |
| $\xi_{\text{FSPL}}(t)$ | Free space path loss                                                                          |
| $\xi_{\text{SF}}$      | Shadow fading                                                                                 |
| $\xi_{\text{CL}}$      | Clutter loss                                                                                  |
| $\xi_c$                | Building entry loss                                                                           |
| $\xi_{\text{rain}}$    | Attenuation caused by atmospheric gasses as well as ionospheric or tropospheric scintillation |
| $f_c$                  | Carrier frequency                                                                             |
| $l_{u,g}(t)$           | Distance between LEO satellite $u$ and GS $g$ in time slot $t$                                |
| $Q_u(t)$               | Queue of LEO satellite $u$ in time slot $t$                                                   |
| $b_u(t)$               | Data amount offloaded from LEO satellite $u$ to one GS in time slot $t$                       |
| $d_{u,g}(t)$           | Transmission time between LEO satellite $u$ and GS $g$ in time slot $t$                       |
| $d_u(t)$               | Transmission time for LEO satellite $u$ to offload the task data in time slot $t$             |
| $\bar{e}_u(t)$         | Energy consumption for LEO satellite $u$ to offload its task data in time slot $t$            |
| $Z_u(t)$               | Virtual queue of LEO satellite $u$ in time slot $t$                                           |
| $e_u^{\text{avg}}$     | Average energy level allowably consumed in LEO satellite $u$                                  |

schemes in LSNs. The system model and problem formulation are then detailed in Section III. Our proposed solutions are introduced in Section IV, followed by simulation results and discussions in Section V. Finally, we give the concluding remarks in Section VI. The main notations used throughout this paper are summarized in Table I.

## II. BACKGROUND AND RELATED WORK

In this section, we first provide an overview of LSN combination applications to highlight the importance of data offloading in LSNs. Subsequently, we discuss the existing literature on the data offloading in this context.

### A. Overview of LSN Combination Applications

The integration of LSNs with various technologies has become a focal point due to their significant potential for enhancing data transmission capabilities. We survey the existing literature

TABLE II  
NETWORK COMBINATION OVERVIEW AND APPLICATION CHARACTERISTICS OF LSNs

| Combination type | Role of LSN                                               | Advantages                                                          | Challenges                                                                                                | Applications                                                           | Key metrics                                                                                                                          |
|------------------|-----------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| LSN+MEC          | Provide low-latency connectivity and data processing [16] | Low latency and high bandwidth                                      | MEC node deployment and dynamic network resource management                                               | Smart cities and Internet of Things (IoT) [17]                         | Energy consumption [18], [19], delay [20]–[22], and network cost [23]–[25]                                                           |
| LSN+Mega         | Provide high-capacity data support                        | Ultra-high speed, ultra-low latency, and comprehensive global reach | Routing design [26]–[28], mobility management [6], handover scheme design [29], and topology control [30] | Video streaming, extensive event broadcasting, and military operations | Energy consumption [26], minimum hop count [27], and end-to-end latency [28]                                                         |
| LSN+SDN          | Support stable communication and network flexibility      | Flexibility and cost efficiency                                     | Control domain division, controller provisioning, and routing design [31]                                 | Low-latency services and video multicasting [32]                       | Total management cost [33], control overhead [34], average service latency [35], network congestion [36], and total path length [37] |
| LSN+Caching      | Enable data caching and content delivery optimization     | Improved data access speed and reduced latency [38]                 | Cache management complexity and data consistency                                                          | Content delivery networks and caching services                         | Delay [39]–[41], energy efficiency [42], and network throughput [43]                                                                 |

on the integration approaches involving LSNs, highlighting their roles, advantages, challenges, applications, and key metrics, summarized in Table II.

*LSN + MEC. Enhancing connectivity with edge computing:* The combination of LSN and Multi-access Edge Computing (MEC) to form a LEO Satellite Edge Computing Network (LSECN) has been extensively studied for its potential to provide low-latency connectivity and data processing capabilities [16]. The LSECN offers low-latency services and high bandwidth, making them ideal for real-time applications such as smart cities and the Internet of Things (IoT) [17]. However, integrating MEC nodes into satellite hardware and managing the dynamic network resources pose significant challenges. To address these issues, numerous studies developed some efficient schemes to optimize the key metrics of LSECN, such as energy consumption [18], [19], delay [20], [21], [22], and network cost [23], [24], [25].

*LSN + Mega. High-capacity data support:* Addressing the escalating requirements for high-capacity data services, LSNs are integrated with mega-constellation networks to establish a robust mega-constellation LSN. This network comprises hundreds to thousands of LEO satellites operating in concert to deliver global broadband coverage [28]. Tailored to offer dependable communication services characterized by ultra-high speed, ultra-low latency, and comprehensive global reach, such network supports applications like video streaming, extensive event broadcasting, and military operations. The extensive scale and multi-hop nature of the mega-constellation LSN pose significant challenges in routing design, particularly concerning convergence and overhead [44]. Consequently, several routing algorithms have been developed for large-scale LEO satellite constellations, focusing on optimizing network metrics such as energy consumption [26], minimum hop count [27], and end-to-end latency [28]. Additionally, the highly dynamic topology of mega-constellation LSN introduces novel challenges in mobility management [6], handover scheme design [29], and topology control [30].

*LSN + SDN. Support stable communication and network flexibility:* Integrating Software-Defined Networking (SDN) into LSN enhances the sustainability and efficiency of satellite networks. SDN facilitates dynamic resource allocation and network management, thereby enhancing flexibility and reducing costs for applications such as low-latency services and video multicasting [32]. However, challenges persist in areas such as control domain division, controller provisioning, and routing design [31]. To this end, a rounding-based dynamic controller assignment strategy [33] has been suggested to minimize total management cost. A controller provisioning strategy [34] with minimal control overhead has been devised, enabling efficient adjustment of satellite controllers using only current network load information. Furthermore, recent research has proposed routing schemes aimed at optimizing metrics including average service latency [35], network congestion [36], and total path length [37].

*LSN + Caching. Optimizing content delivery:* The low latency and high bandwidth of LEO satellites in LSN also make them suitable for content delivery optimization through caching mechanisms. By caching popular content on LEO satellites, data access speed can be significantly improved, reducing latency and enhancing user experience for applications such as content delivery networks [38]. However, the dynamic nature of the satellite environment introduces significant challenges, particularly in managing cache consistency and addressing the complexities associated with cache management. To effectively tackle these challenges, various caching strategies have been devised to either minimize key performance metrics, such as link delay [39], communication delay [40], and average content retrieval delay [41], or to maximize energy efficiency [42] and network throughput [43].

In addition to the aforementioned technologies, there are several other notable advancements, such as digital twin [45] and AI. These include deep reinforcement learning [46], graph optimization and learning [47], [48], and generative diffusion models [49], among others. The integration of LSNs with these

technologies underscores their critical role in enhancing data transmission capabilities. While each integration approach faces unique challenges, the role of LSNs in guaranteeing reliable, low-latency, and high-bandwidth communication is of utmost importance. This underscores the critical importance of data offloading within LSNs, as it serves as the cornerstone for attaining the key performance metrics.

### B. Data Offloading in LSNs

Recent research has extensively studied the data offloading problems in LSNs. We categorize them into two primary categories: the offline and online data offloading schemes.

1) *Offline Data Offloading Schemes in LSNs*: Existing offline data offloading schemes for LSNs address two major challenges to improve the efficiency of data offloading. The first challenge is the characterization of the line-of-sight communication links in LSNs [50]. The second challenge lies in discovering new optimization dimensions to enhance the offloading capacity of LSNs. To address the initial challenge, several studies [51], [52], [53], [54] adopt the time window model to depict the inter-satellite links between LEO satellites in LSNs and Geosynchronous (GEO) satellites in Data Relay Satellites Networks (DRSNs). Notably, these works integrate time window constraints into their problem formulations. Consequently, their problems belong to a general class of parallel machine scheduling with time windows, which is known to be NP-hard [70]. For practical considerations, they focus on designing efficient approximation algorithms [51], [52], [53], [54] to improve the data offloading performance. The second category of research employs the graph models, such as the snapshot graph [55] and the time-expanded graph [56], [57], to depict the dynamic communication links of LSNs. These studies [55], [56], [57] primarily use the graph models as mathematical tools to facilitate problem formulation or resolution related to data offloading in LSNs. To address the second challenge, recent studies have proposed new optimization dimensions of multiple satellite collaboration [58], [59] and joint optimization of multiple resources [60], [61], [62] to boost the performance of LSNs. For instance, Jia et al. [58] exploited Inter-Satellite Links (ISLs) to enable collaboration among multiple LEO satellites for maximizing data offloading throughput in LSNs. In view of this, our prior work [59] proposed a collaborative data offloading scheme for Earth Observation Satellite Networks (EOSNs). The works of [60], [61], [62] jointly optimized the observation and communication resources in EOSNs to maximize the sum priorities of the successfully scheduled tasks. However, the aforementioned works [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62] primarily focus on *offline* data offloading strategies. Therefore, these strategies are not applicable for handling the stochastic arrivals of space data and the time-varying network resources.

2) *Online Data Offloading Schemes in LSNs*: Some state-of-the-art works studied the *online* data offloading problems, aiming to capture the stochastic arrivals of space data [63], [64], [65]. However, these dynamic scheduling schemes [63], [64], [65] are not apt for our studied problem due to the following

reasons. Firstly, these dynamic scheduling schemes are tailored for DRSNs to real-timely schedule ISLs in a relatively stable space environment. As such, they do not accommodate the allocation of time-varying SGLs of LSNs in a changing low-space environment. Secondly, their scheduling modes operate on large time-scales in the form of data blocks rather than packets, leading to scheduling time-scales that are too large for the online requirements of LSNs. Lastly, these schemes do not account for the energy consumption of LEO satellites, which significantly impacts the data offloading performance of LSNs. To learn efficient offloading policies from the dynamic environments of LSNs, Cheng et al. jointly optimized task scheduling and resource allocation to devise a Deep Actor Critic (DAC) based algorithm [66]. The proposed DAC-based algorithm is to minimize the delay cost and the energy consumption cost under the constraints of the maximum delay and computation resources. To cope with the unknown environment dynamics of LSNs, Gao et al. [67] proposed a novel online scheduling mechanism that jointly leveraged online learning and online control to determine both the task scheduling and CPU cycle frequency allocation with the historical information. The proposed online scheduling mechanism is to minimize the total task latency while satisfying the constraints of time-averaged energy consumption. Different from [67], Li et al. [68] developed a dual-network deep reinforcement learning offloading scheme to minimize both system delay and energy consumption. In addition, Huang et al. [69] developed a dynamic power allocation strategy for the time-varying uplink from IoT gateways to LEO satellites, to minimize energy consumption and maximize the uploaded data amount for GEO-distributed IoT networks.

However, the assumption in the schemes of [66], [67], [68], [69] that network metrics, such as task latency, energy consumption, and data amount, can be immediately obtained once the offloading decisions are made, may be overly optimistic. This is because the time-varying SGLs in LSNs may cause a large delay in the feedback of network metrics for several time slots. In Table III, we succinctly outline the distinctions between existing works on data offloading for LSNs and our work. In view of analyzing the existing works, it is significant to jointly optimize power allocation and task scheduling for LSNs with considering the energy consumption to accommodate the stochastic space data arrivals and time-varying SGLs.

### III. SYSTEM MODEL AND PROBLEM FORMULATION

Consider a time-slotted LSN consisting of a set  $\mathcal{U} = \{1, 2, \dots, U\}$  of LEO satellites and a set  $\mathcal{G} = \{1, 2, \dots, G\}$  of GSs. In the LSN, the LEO satellites are required to offload the data they carry to the set of GSs, as shown in Fig. 1. Here, we assume that each LEO satellite is outfitted with a single data transmit antenna, while each GS is equipped with an array of data receiver antennas. The scheduling horizon is divided into a set of time slots, denoted by  $\mathcal{T} = \{1, 2, \dots, T\}$ . Each time slot  $t \in \mathcal{T}$  indexes the time segment of  $[t, t + 1]$  with the same duration of  $\tau$ . The primary characteristic of communication between LEO satellites and GSs is the occurrence of intermittent SGLs. In this paper, only a single-hop path exists between a LEO satellite

TABLE III  
COMPARISON OF EXISTING WORKS

|                         | [51]–[54] | [55]–[62] | [63] | [64] | [65] | [66] | [67] | [68] | [69] | Our work |
|-------------------------|-----------|-----------|------|------|------|------|------|------|------|----------|
| SGLs                    | ×         | ✓         | ×    | ×    | ×    | ✓    | ✓    | ✓    | ✓    | ✓        |
| Online scheme           | ×         | ×         | ✓    | ✓    | ✓    | ✓    | ✓    | ✓    | ✓    | ✓        |
| Power allocation        | ×         | ×         | ×    | ×    | ×    | ×    | ✓    | ✓    | ✓    | ✓        |
| Task scheduling         | ✓         | ✓         | ✓    | ✓    | ✓    | ✓    | ✓    | ✓    | ✓    | ✓        |
| Time-varying SGLs       | ×         | ×         | ×    | ×    | ×    | ×    | ×    | ×    | ×    | ✓        |
| Bounded performance gap | ×         | ×         | ×    | ×    | ×    | ×    | ✓    | ×    | ×    | ✓        |

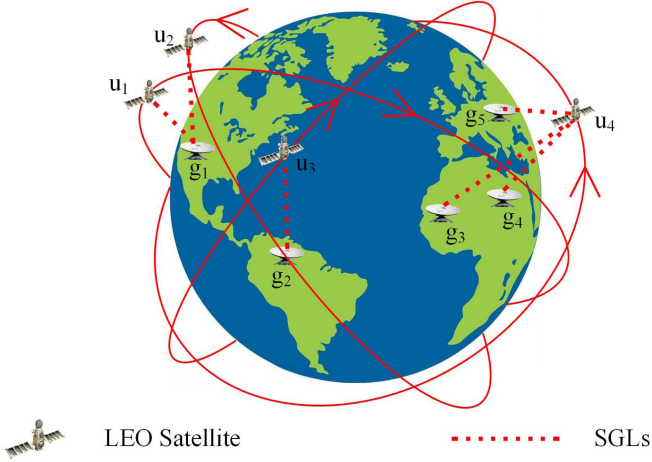


Fig. 1. The network scenario of time-slotted LSN.

and a GS, contrasting with the multi-hop dynamic topology. Therefore, we solely utilize  $\mathcal{G}_u(t)$  to represent the subset of GSs that can establish a connection with LEO satellite  $u$  in each time slot  $t \in \mathcal{T}$ .

#### A. Task Model and Channel Model

We consider a task model with sequential dependencies, where a new task is generated at the beginning of each time slot and the task arrivals are independent and identically distributed across time slots. Denote  $a_u(t)$  as the data amount for the task generated on-board LEO satellite  $u$  in time slot  $t \in \mathcal{T}$ . We denote  $\mathbf{a}(t) = \{a_u(t)\}$  as the set of data amounts of all LEO satellites in time slot  $t \in \mathcal{T}$ . We introduce a binary variable  $x_{u,g}(t) \in \{0, 1\}$  to represent the data offloading strategy, such that  $x_{u,g}(t) = 1$  indicates that LEO satellite  $u \in \mathcal{U}$  selects the GS  $g \in \mathcal{G}_u(t)$  to offload task data in the time slot  $t \in \mathcal{T}$ ; and  $x_{u,g}(t) = 0$  otherwise.

Let  $R_{u,g}(t)$  denote the achievable data rate of SGL  $(u, g)$  in time slot  $t$ , expressed as

$$R_{u,g}(t) = B_c \log_2(1 + \gamma_{u,g}(t)), \forall g \in \mathcal{G}_u(t), t \in \mathcal{T}, \quad (1)$$

where  $B_c$  is the dedicated bandwidth and  $\gamma_{u,g}(t)$  is the signal-to-noise ratio of SGL  $(u, g)$  in time slot  $t$ . In particular,  $\gamma_{u,g}(t)$  can be evaluated by

$$\gamma_{u,g}(t) = \frac{\Psi_{u,g}(t)P_{u,g}(t)}{\sigma_g^2(t)}, \forall g \in \mathcal{G}_u(t), t \in \mathcal{T}. \quad (2)$$

In (2),  $P_{u,g}(t)$  is the allocated transmit power (in Watts) of SGL  $(u, g)$  in time slot  $t$ ,  $\sigma_g^2(t)$  is the noise power at GS  $g$  in time slot

$t$ , and  $\Psi_{u,g}(t)$  is the channel gain of SGL  $(u, g)$  in time slot  $t$ , calculated by

$$\Psi_{u,g}(t) = 10^{-\xi_{u,g}(t)/10} h_u^{\text{tr}}(t) h_g^{\text{re}}(t). \quad (3)$$

Wherein,  $h_u^{\text{tr}}(t)$  is the transmit antenna gain of LEO satellite  $u$  in time slot  $t$ ,  $h_g^{\text{re}}(t)$  is the gain of data receiver antenna on GS  $g$  in time slot  $t$ , and  $\xi_{u,g}(t)$  is the path loss (in dB) between LEO satellite  $u$  and GS  $g$  in time slot  $t$ . The path loss  $\xi_{u,g}(t)$  is given by

$$\xi_{u,g}(t) = \xi_{\text{FSPL}}(t) + \xi_{\text{SF}} + \xi_{\text{CL}} + \xi_e + \xi_{\text{rain}}, \quad (4)$$

where  $\xi_{\text{FSPL}}(t) = 32.45 + 20 \log_{10}(f_c) + 20 \log_{10}(l_{u,g}(t))$  is the free space path loss,  $\xi_{\text{SF}}$  is the shadow fading following a zero-mean normal distribution with a standard deviation  $\sigma_{\text{SF}}^2$ , i.e.,  $N(0, \sigma_{\text{SF}}^2)$ ,  $\xi_{\text{CL}}$  is clutter loss,  $\xi_e$  is building entry loss, and  $\xi_{\text{rain}}$  is the attenuation caused by atmospheric gasses as well as ionospheric or tropospheric scintillation [71]. The notation  $f_c$  is the carrier frequency and  $l_{u,g}(t)$  denotes the distance between LEO satellite  $u$  and GS  $g$  in time slot  $t$ . We assume that  $\Psi_{u,g}(t)$  is limited to the range  $[\Psi_{\min}, \Psi_{\max}]$ , where  $\Psi_{\min}$  and  $\Psi_{\max}$  are the minimum and maximum values, respectively. The channel model captures the characteristics of SGLs via key components like distance, frequency, environment, and antenna properties, which account for physical phenomena and practical aspects in LSNs. This enables realistic communication performance assessment and effective resource allocation strategy development.

#### B. Queue Update and Energy Consumption Model

The arrival and departure procedure of task data in any LEO satellite  $u$  for each time slot  $t$  is modeled as a queue updating process. This queue denoted by  $Q_u(t)$  is dynamically updated over time slots according to the following rule:

$$Q_u(t+1) = \max[Q_u(t) - b_u(t), 0] + a_u(t), \forall u, t, \quad (5)$$

where  $b_u(t)$  is the data amount offloaded from LEO satellite  $u$  to one GS in the time slot  $t$ , given by

$$b_u(t) = R_u(t) d_u(t), \forall t. \quad (6)$$

In (6),  $R_u(t)$  is the offload transmission rate of LEO satellite  $u$  in time slot  $t$ , expressed as

$$R_u(t) = \sum_{g \in \mathcal{G}_u(t)} x_{u,g}(t) R_{u,g}(t), \forall t, \quad (7)$$

and  $d_u(t)$  indicates the transmission time for LEO satellite  $u$  to offload the task data in time slot  $t$ , evaluated by

$$d_u(t) = \sum_{g \in \mathcal{G}_u(t)} x_{u,g}(t) d_{u,g}(t), \forall t. \quad (8)$$

In (8),  $d_{u,g}(t)$  represents the transmission time between LEO satellite  $u$  and GS  $g$  within time slot  $t$ , given by

$$d_{u,g}(t) = \begin{cases} \frac{Q_u(t)}{R_{u,g}(t)} & Q_u(t) < R_u(t)\tau, \\ \tau & \text{otherwise.} \end{cases} \quad (9)$$

Let  $e_u(t)$  denote the energy consumption for LEO satellite  $u$  to offload its task data in time slot  $t$ , calculated by

$$e_u(t) = \sum_{g \in \mathcal{G}_u(t)} x_{u,g}(t) P_{u,g}(t) d_{u,g}(t), \forall u, t. \quad (10)$$

### C. Joint Online Optimization Problem Formulation

By jointly optimizing power allocation and task scheduling, we formulate an online data offloading problem for LSNs to maximize the data amount of offloaded tasks while prolonging the battery lifetime of satellites. This leads to the following joint online optimization problem:

$$\begin{aligned} \mathbf{P0}: & \max_{\mathbf{P}(t), \mathbf{x}(t)} \lim_{|\mathcal{T}| \rightarrow \infty} \frac{1}{|\mathcal{T}|} \sum_{u \in \mathcal{U}} \sum_{t \in \mathcal{T}} \mathbb{E} \{b_u(t)\} \\ \text{s.t. } & \mathbf{C1} : \lim_{|\mathcal{T}| \rightarrow \infty} \sup_{t \in \mathcal{T}} \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathbb{E} \{e_u(t)\} \leq e_u^{\text{avg}}, \forall u, \\ & \mathbf{C2} : P_{\min} \leq P_{u,g}(t) \leq P_{\max}, \forall u, t, g \in \mathcal{G}_u(t), \\ & \mathbf{C3} : \sum_{g \in \mathcal{G}_u(t)} x_{u,g}(t) \leq 1, \forall u, t, \\ & \mathbf{C4} : x_{u,g}(t) \in \{0, 1\}, \forall u, t, g \in \mathcal{G}_u(t), \\ & \mathbf{C5} : \text{Queue } Q_u(t) \text{ is mean rate stable}, \forall u, t, \end{aligned}$$

where  $\mathbf{P}(t) = \{P_{u,g}(t)\}$ ,  $\mathbf{x}(t) = \{x_{u,g}(t)\}$ ,  $P_{\min}$  is the minimum transmit power,  $P_{\max}$  is the maximum transmit power, and  $e_u^{\text{avg}}$  indicates the average energy level allowably consumed in LEO satellite  $u$ . C1 indicates that the energy consumption of any LEO satellite cannot exceed its average level of allowable energy consumption from the long-term average perspective. C2 indicates the power control range of any LEO satellite. C3 reflects that any LEO satellite selects at most one GS in each time slot. C4 is the binary constraint. C5 is the constraint of the mean rate queue stability to ensure a finite queue length of any LEO satellite.

*Remark 1:*  $\mathbf{P0}$  is challenging to solve due to the following aspects: First, it is intractable to real-timely acquire the accurate channel information of  $\Psi_{u,g}(t)$  in LSNs, since it is a time-varying and time-delayed random variable. Second, the unpredictable data amount of arrival tasks  $\mathbf{a}(t)$  significantly degrades the data offloading efficiency of LSNs. Third, the decision variables of  $\mathbf{P}(t)$  and  $\mathbf{x}(t)$  are coupled in both C1 and the objective function, thereby greatly increasing the solving difficulty of  $\mathbf{P0}$ . Finally, both the objective function and C1 in  $\mathbf{P0}$  are long-time average, resulting in the failure of the traditional optimization tools.

## IV. PROPOSED SOLUTIONS

In this section, we first provide an overview of the problem solving framework for  $\mathbf{P0}$ . Then, we study the simplified version

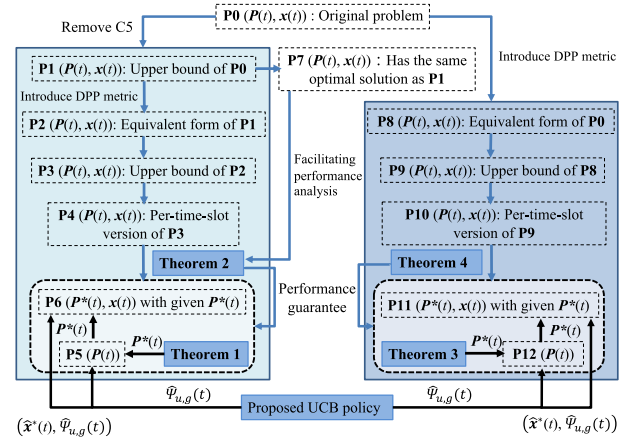


Fig. 2. Solving framework for the studied problem.

of  $\mathbf{P0}$  that neglects the constraint of mean rate queue stability, to propose an efficient online solution. Finally, we extend the special solution to solve the original problem  $\mathbf{P0}$ .

### A. Solving Framework Overview

The brief overview of the problem solving framework for  $\mathbf{P0}$  is shown in Fig. 2. Specifically, we first remove C5 from  $\mathbf{P0}$  to obtain its simplified version, i.e.,  $\mathbf{P1}$ . Then, we introduce the Drift-Plus-Penalty (DPP) metric to equivalently transform  $\mathbf{P1}$  into  $\mathbf{P2}$ . Since  $\mathbf{P2}$  is non-convex, we derive the upper problem  $\mathbf{P3}$  by using relaxation operation. By exploiting the special structure of  $\mathbf{P3}$ , we decompose it into a series of per-time-slot optimization subproblems, each represented by  $\mathbf{P4}$ . Next, we adopt a two-level optimization method to divide  $\mathbf{P4}$  into a power allocation problem  $\mathbf{P5}$  and a task scheduling problem  $\mathbf{P6}$ . We further provide Theorem 1 to show a closed-form optimal solution to  $\mathbf{P5}$ . Then, we combine the MAB framework with the Upper Confidence Bound (UCB) policy [67] to solve  $\mathbf{P6}$ . Furthermore, we summarize the above solutions to solve  $\mathbf{P1}$  in Algorithm 1. To facilitate the performance analysis, we transform  $\mathbf{P1}$  into  $\mathbf{P7}$  and then present Theorem 2 to show that the performance gap with optimal solution to  $\mathbf{P1}$  is bounded. Following the aforementioned solving framework, we can efficiently address  $\mathbf{P0}$  through the transformation and decomposition process that transitions from  $\mathbf{P8}$  to  $\mathbf{P12}$ . Furthermore, we present Algorithm 2 to address problem  $\mathbf{P0}$ . Finally, we provide Theorem 3 and Theorem 4 to demonstrate a closed-form optimal solution for  $\mathbf{P12}$  and to quantify the performance loss of Algorithm 2, respectively.

### B. Online Data Offloading W.o. Mean Rate Queue Stability

1) *Virtual Queue and Per-Slot Optimization Problem:* Without considering the constraint of mean rate queue stability, we obtain the following simplified form of  $\mathbf{P0}$ :

$$\begin{aligned} \mathbf{P1}: & \max_{\mathbf{P}(t), \mathbf{x}(t)} \lim_{|\mathcal{T}| \rightarrow \infty} \frac{1}{|\mathcal{T}|} \sum_{u \in \mathcal{U}} \sum_{t \in \mathcal{T}} \mathbb{E} \{b_u(t)\} \\ \text{s.t. } & \mathbf{C1-C4}. \end{aligned}$$

To handle constraint C1, we introduce a virtual queue  $Z_u(t)$  for any LEO satellite  $u$  with the following updating rule:

$$Z_u(t+1) = \max\{Z_u(t) + e_u(t) - e_u^{\text{avg}}, 0\}, \forall u, t. \quad (11)$$

We further introduce the *Lyapunov function* of  $L_1(Z_u(t)) = \frac{1}{2}Z_u(t)^2$  to convert C1 into the stability of virtual queues. On this basis, we provide the definition of *conditional Lyapunov drift* for each time slot to show the change in the *Lyapunov function* from one slot to the next as follows:

$$\Delta_1(Z_u(t)) = \mathbb{E}\{L_1(Z_u(t+1)) - L_1(Z_u(t)) | Z_u(t)\}, \forall u, t.$$

As such, the stability of virtual queue  $Z_u(t)$  can be equivalently transformed into minimizing  $\Delta_1(Z_u(t))$  [72]. Furthermore, we introduce a DPP metric to transform P1 into the following problem:

$$\mathbf{P2}: \min_{P(t), x(t)} \lim_{|\mathcal{T}| \rightarrow \infty} \frac{1}{|\mathcal{T}|} \sum_{\substack{t \in \mathcal{T} \\ u \in \mathcal{U}}} (\Delta_1(Z_u(t)) - V\mathbb{E}\{b_u(t)\})$$

s.t. C2-C4,

where  $V$  is a positive parameter to balance the stability of virtual queue  $Z_u(t)$  and the offloaded data amount. Directly solving P2 is still difficult, since its objective function is non-convex. As such, we turn to calculating an upper bound of the optimal objective value of P2. Using the definitions of  $L_1(Z_u(t))$  and  $\Delta_1(Z_u(t))$ , we have  $\Delta_1(Z_u(t)) - V\mathbb{E}\{b_u(t)\} \leq \mathbb{E}\{Z_u(t)(e_u(t) - e_u^{\text{avg}}) | Z_u(t)\} - V\mathbb{E}\{b_u(t)\} + B_1$ , where  $B_1 = \frac{1}{2}(P_{\max}\tau - e_u^{\text{avg}})^2$ . Then, we obtain an upper bound problem of P2 as follows:

$$\mathbf{P3}: \min_{P(t), x(t)} \lim_{|\mathcal{T}| \rightarrow \infty} \frac{1}{|\mathcal{T}|} \sum_{\substack{t \in \mathcal{T} \\ u \in \mathcal{U}}} \left( \mathbb{E}\{Z_u(t)(e_u(t) - e_u^{\text{avg}}) | Z_u(t)\} - V\mathbb{E}\{b_u(t)\} \right) + B_1$$

s.t. C2-C4.

Furthermore, we remove the constant term  $B_1$  from P3 and transform it into the following per-time-slot optimization subproblem for any LEO satellite  $u$  at any time slot  $t$ :

$$\mathbf{P4}: \min_{P(t), x(t)} \sum_{g \in \mathcal{G}} x_{u,g}(t) \mathfrak{S}_{u,g}(t)$$

s.t. C6 :  $P_{\min} \leq P_{u,g}(t) \leq P_{\max}, \forall g \in \mathcal{G}_u(t)$ ,

$$\text{C7 : } \sum_{g \in \mathcal{G}_u(t)} x_{u,g}(t) \leq 1,$$

$$\text{C8 : } x_{u,g}(t) \in \{0, 1\}, \forall g \in \mathcal{G}_u(t),$$

where  $\mathfrak{S}_{u,g}(t) = (Z_u(t)P_{u,g}(t) - VR_{u,g}(t))d_{u,g}(t)$ .

2) *Solution for Per-Slot Optimization Subproblem:* We separate P4 into two levels of optimization. At the lower level of optimization, we optimize  $P(t)$  with fixed  $x(t)$  and known  $\Psi_{u,g}(t)$ :

$$\mathbf{P5}: \min_{P(t)} \mathfrak{S}_{u,g}(t)$$

s.t. C6.

At the higher level of optimization, we optimize  $x(t)$  when fixing  $P(t)$ :

$$\mathbf{P6}: \min_{x(t)} \sum_{g \in \mathcal{G}} x_{g,t} \mathfrak{S}_{u,g}^*(t)$$

s.t. C7-C8,

where  $\mathfrak{S}_{u,g}^*(t)$  is an optimal value to P5.

Next, we devote to successively solving problems P5 and P6. In detail, we first obtain the two extremal points of P5 as

$$P_0^{u,g}(t) = \frac{\sigma_g^2(t)}{\Psi_{u,g}(t)} \left( 2^{\frac{Q_{u,g}(t)}{\tau B_c}} - 1 \right), \quad (12)$$

and

$$P_{u,g,1}^0(t) = \frac{VB_c}{Z_u(t) \ln 2} - \frac{\sigma_g^2(t)}{\Psi_{u,g}(t)}. \quad (13)$$

If  $P_{u,g}(t) \in [P_{\min}, P_0^{u,g}(t)]$ , we further obtain an optimal solution for minimizing  $\mathfrak{S}_{u,g}(t)$  as follows:

$$P_1^{u,g}(t) = \begin{cases} P_0^{u,g}(t) & Z_u(t) < \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_0^{u,g}(t)) \ln 2}, \\ P_{\min} & Z_u(t) > \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\min}) \ln 2}, \\ P_{u,g,1}^0(t) & \text{otherwise.} \end{cases} \quad (14)$$

For the convenient presentation of the optimal solution to P5, we require to define the following functions as

$$\Phi_{P_{u,g}(t)}^1 = \log_2 \left( \frac{2Z_u(t)P_{u,g}(t)}{\left(1 + \frac{\Psi_{u,g}(t)}{\sigma_g^2(t)}P_{u,g}(t)\right)^{VB_c}} \right), \quad (15)$$

and

$$\varphi_{\eta_{u,g}(t)}^1 = \frac{\sigma_g^2(t)Z_u(t)}{\Psi_{u,g}(t)} \eta_{u,g}(t) \left( 2^{\frac{1}{B_c \eta_{u,g}(t)}} - 1 \right) - V, \quad (16)$$

where  $\eta_{u,g}(t) = \chi(P_{u,g}(t))$  is a new variable used to replace  $P_{u,g}(t)$  and the function of  $\chi(\cdot)$  is defined as

$$\chi(P_{u,g}(t)) = \frac{1}{B_c \log_2 \left( 1 + \frac{\Psi_{u,g}(t)}{\sigma_g^2(t)}P_{u,g}(t) \right)}. \quad (17)$$

Theorem 1 below provides a closed-form optimal solution for problem P5.

*Theorem 1:* The optimal solution  $P_{u,g}^*(t)$  to P5 is

$$P_{u,g}^*(t) = \begin{cases} P_{u,g,1}^*(t) & P_0^{u,g}(t) \in [P_{\min}, P_{\max}], \\ P_{u,g,2}^*(t) & P_0^{u,g}(t) \in (P_{\max}, \infty), \\ P_{\min} & P_0^{u,g}(t) \in (0, P_{\min}), \end{cases} \quad (18)$$

where  $P_{u,g,1}^*(t)$  and  $P_{u,g,2}^*(t)$  are respectively defined as

$$P_{u,g,1}^*(t) = \begin{cases} P_1^{u,g}(t) & \frac{\Phi_{P_1^{u,g}(t)}^1}{Q_{u,g}(t)} \leq \frac{\varphi_{\chi(P_1^{u,g}(t))}^1}{\tau}, \\ P_0^{u,g}(t) & \text{otherwise,} \end{cases} \quad (19)$$

and

$$P_{u,g,2}^*(t) = \begin{cases} P_{\max} & Z_u(t) < \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\max}) \ln 2}, \\ P_{\min} & Z_u(t) > \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\min}) \ln 2}, \\ P_{u,g,1}^0(t) & \text{otherwise.} \end{cases} \quad (20)$$

*Proof:* See Appendix A. ■

The main steps for the proof of Theorem 1 are summarized as follows: First, we study the location relationship between the value of the extremal point  $P_{u,g}^0(t)$  of  $\mathfrak{S}_{u,g}(t)$  and the power allocation range  $[P_{\min}, P_{\max}]$ . In particular, we consider three cases: one with  $P_{u,g}^0(t) \in [P_{\min}, P_{\max}]$ ; one with  $P_{u,g}^0(t)$  falling to the left of the interval  $[P_{\min}, P_{\max}]$ ; and one with  $P_{u,g}^0(t)$  located to the right of  $[P_{\min}, P_{\max}]$ . Furthermore, we use convex optimization methods to obtain the optimal power allocation  $P_{u,g}^*(t)$  for the three cases.

*Remark 2:* Theorem 1 implies that the key factors that influence  $P_{u,g}^*(t)$  are the queue value  $Z_u(t)$  and the parameter  $V$ . When the value of  $Z_u(t)$  is large, a minimal power value, such as  $P_{\min}$ , is allocated to maintain constraint C1. In contrast, when  $Z_u(t)$  is small, a larger value, such as  $P_{u,g,1}^0(t)$  or potentially  $P_{\max}$ , is adaptively allocated to minimize the objective value of **P5**. Similarly, an increase in the value of  $V$  results in a higher power value,  $P_{u,g,1}^0(t)$  or potentially  $P_{\max}$ , being allocated to  $P_{u,g}^*(t)$ . Conversely, a lower power value  $P_{\min}$  is allocated to  $P_{u,g}^*(t)$ .

The time-varying and time-delayed SGLs bring two main challenges in solving **P6**: First, it is hard to obtain  $\mathfrak{S}_{u,g}^*(t)$  due to the difficulty in accurately calculating the channel gains of  $\Psi_g(t)$  for all  $g$  and  $t$ . Second,  $\mathfrak{S}_{u,g}^*(t)$  cannot be immediately obtained after making decision  $x_{g,t}$ . To address these challenges, we adopt MAB framework with UCB policy [67] to tackle **P6**. In detail, we treat each GS as an arm, each associated with a cost unknown beforehand, i.e., the objective value of **P6**. As such, **P6** can be equivalently reformulated as a MAB problem to minimize the cost in each time slot through the arm selection. Furthermore, we leverage the UCB policy to learning the unknown yet delayed  $\Psi_g(t)$ ,  $\forall g, t$ . Directly applying the UCB policy to predict the channel gain of  $\Psi_{u,g}(t)$  using the traditional method in [67] suffers from significant performance degradation. This is primarily due to the fact that the path loss  $\xi_{u,g}(t)$ , as defined in (4), is influenced by both the time-varying distance  $l_{u,g}(t)$  and the shadow fading  $\xi_{SF}$ . Therefore, it is challenging to effectively learn time-varying feature of  $l_{u,g}(t)$  and randomness of  $\xi_{SF}$  to directly predict  $\hat{\Psi}_{u,g}(t)$  with high accuracy. In contrast to traditional UCB methods, we propose to estimate  $\xi_{SF}$  first and then calculate  $\hat{\Psi}_{u,g}(t)$  so as to avoid the distortion caused by time-varying distance  $l_{u,g}(t)$ , thus significantly improving the prediction accuracy of channel gain  $\Psi_{u,g}(t)$ .

The main steps of the proposed UCB policy are given as follows. First, we need to obtain an estimation of the channel gain of SGL  $(u, g)$  in time slot  $t$ :

$$\hat{\Psi}_{u,g}(t) = \bar{\Psi}_{u,g}(t-1) + \sqrt{\frac{3 \ln t}{2n_{u,g}(t-1)}}. \quad (21)$$

In (21),  $\bar{\Psi}_{u,g}(t)$  represents the sample mean of  $\Psi_{u,g}(t)$ , i.e.,

$$\bar{\Psi}_{u,g}(t) = \begin{cases} \frac{1}{n_{u,g}(t)} \sum_{k=1}^t x_{u,g}(k) \Psi_{u,g}(k) & n_{u,g}(t) > 0, \\ 0 & n_{u,g}(t) = 0, \end{cases} \quad (22)$$

where  $n_{u,g}(t)$  represents the number of selected times of SGL  $(u, g)$  until the end of time slot  $t$ , i.e.,

$$n_{u,g}(t) = \sum_{k=1}^t x_{u,g}(k), \forall t. \quad (23)$$

Particularly, we set  $\hat{\Psi}_{u,g}(t) = 0$  when  $n_{u,g}(t-1) = 0$ . Second, we use  $\hat{\Psi}_{u,g}(t)$  to acquire the estimated transmission rate for GS  $g$  in time slot  $t$ , denoted by  $\hat{R}_{u,g}(t)$ . Specifically, we combine (1) with (2) to calculate the value of  $\hat{R}_{u,g}(t)$ , i.e.,  $\hat{R}_{u,g}(t) = B_c \log_2(1 + \frac{\hat{\Psi}_{u,g}(t)}{\sigma_g^2(t)} \hat{P}_{u,g}^*(t))$ , where  $\hat{P}_{u,g}^*(t)$  denotes the optimal power solution to **P5** based on  $\hat{\Psi}_{u,g}(t)$ ,  $\forall g \in \mathcal{G}_u(t)$ . Third, we use  $\hat{R}_{u,g}(t)$  to calculate the estimated objective function value of **P5**, which is equal to  $\sum_{g \in \mathcal{G}} x_{u,g}(t) (Z_u(t) \hat{P}_{u,g}^*(t) - V \hat{R}_{u,g}(t)) \hat{d}_{u,g}(t)$  with  $\hat{d}_{u,g}(t) = \min(\tau, \frac{Q_u(t)}{\hat{R}_{u,g}(t)})$  denoting the estimated transmission time between LEO satellite  $u$  and GS  $g$ . Furthermore, we use the UCB policy to yield  $\hat{x}^*(t)$  based on  $\hat{\Psi}_{u,g}(t)$ ,  $\forall g \in \mathcal{G}_u(t)$ ,  $u, t$ , satisfying that  $\hat{x}_{u,g}^*(t) = 1$  when

$$g^* = \arg \min_{g \in \mathcal{G}_u(t)} (Z_u(t) \hat{P}_{u,g}^*(t) - V \hat{R}_{u,g}(t)) \hat{d}_{u,g}(t), \quad (24)$$

and otherwise  $\hat{x}_{u,g}^*(t) = 0$ . Finally, we summarize the main steps of solving **P1** in Algorithm 1, termed Energy-Constrained Online Data Offloading algorithm with Single Queue (EODO-SQ).

To quantify the performance loss of EODO-SQ, we define the time-averaged regret as follows:

$$\text{Reg}(T) = \frac{1}{T} \sum_{u \in \mathcal{U}} \sum_{t \in \mathcal{T}} (R_u^*(t) d_u^*(t) - \mathbb{E}\{b_u(t)\}), \quad (25)$$

where  $R_u^*(t)$  and  $d_u^*(t)$  denote the offload transmission rate and the transmission time of the theoretical optimal solution to **P1**, respectively. On this basis, **P1** can be equivalently transformed into the regret minimization problem below:

$$\begin{aligned} \mathbf{P7}: \quad & \min_{\mathbf{P}(t), \mathbf{x}(t)} \text{Reg}(T) \\ & \text{s.t. C1-C4.} \end{aligned}$$

Theorem 2 shows that the performance gap between EODO-SQ and the optimal solution to **P7** is bounded.

*Theorem 2:* The theoretical upper bound for the time-averaged regret achieved by EODO-SQ is

$$\text{Reg}(T) \leq \frac{B_1 U}{V} + \frac{\tau B_c P_{\max} U}{T \sigma_{\min}^2 \ln 2} F(T), \quad (26)$$

where  $F(T) = (2G + 1)(\Psi_{\max} - \Psi_{\min}) + 2\sqrt{6GT \ln T}$ .

*Proof:* See Appendix B. ■

The proof of Theorem 2 mainly includes the following steps: Initially, we employ the DPP metric to establish a preliminary bound for  $\text{Reg}(T)$ , followed by transforming **P7** into a series of

**Algorithm 1:** Proposed EODO-SQ Algorithm.

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**Input:**  $\mathcal{G}_u(t)$ ,  $P_{\min}$ ,  $P_{\max}$ ,  $\tau$ ,  $B_c$ ,  $e_u^{\text{avg}}$ ,  $V$ .

- 1: **Initialization:**  $\bar{\Psi}_{u,g}(0) = 0$ ,  $n_{u,g}(0) = 0$ ,  $Q_u(0) = 0$ ,  $Z_u(0) = 0$ .
- 2: **for**  $t = 1, \dots, T$  **do**
- 3:   **for**  $u = 1, \dots, U$  **do**
- 4:     **for**  $g = 1, \dots, G$  **do**
- 5:       **if**  $g \in \mathcal{G}_u(t)$  **then**
- 6:          Use (21) to compute  $\hat{\Psi}_{u,g}(t)$ .
- 7:          Use (12) to obtain  $P_{0,g}^{u,g}(t)$ .
- 8:          Use (13) to calculate  $P_{u,g,1}^0(t)$ .
- 9:          Use (14) to compute  $P_{1,g}^{u,g}(t)$ .
- 10:        Use (15) to obtain  $\Phi_{P_{1,g}^{u,g}(t)}^1$ .
- 11:        Use (16) and (17) to compute  $\varphi_{\chi(P_{0,g}^{u,g}(t))}^1$ .
- 12:        **if**  $P_{0,g}^{u,g}(t) \in [P_{\min}, P_{\max}]$  **then**
- 13:          **if**  $\Phi_{P_{1,g}^{u,g}(t)}^1 \tau \leq \varphi_{\chi(P_{0,g}^{u,g}(t))}^1 Q_u(t)$  **then**
- 14:             $\hat{P}_{u,g}^*(t) = P_{1,g}^{u,g}(t)$ .
- 15:          **else**
- 16:             $\hat{P}_{u,g}^*(t) = P_{0,g}^{u,g}(t)$ .
- 17:          **end if**
- 18:        **else if**  $P_{0,g}^{u,g}(t) \in (P_{\max}, \infty)$  **then**
- 19:          Use (20) to obtain  $\hat{P}_{u,g}^*(t)$ .
- 20:        **else**
- 21:           $\hat{P}_{u,g}^*(t) = P_{\min}$ .
- 22:        **end if**
- 23:        **end if**
- 24:        **end for**
- 25:        Use (24) to obtain  $\hat{x}_{u,g}^*(t)$ .
- 26:        Use (5) and (11) to update  $Q_u(t)$  and  $Z_u(t)$ .
- 27:        Use (22) and (23) to update  $\bar{\Psi}_{u,g}(t)$  and  $n_{u,g}(t)$ .
- 28:        **end for**
- 29:        **end for**

**Output:**  $\hat{P}^*(t) = \{\hat{P}_{u,g}^*(t)\}$ ,  $\hat{x}^*(t) = \{\hat{x}_{u,g}^*(t)\}$ .

---

per-time-slot subproblems in order to determine an explicit upper bound for  $\text{Reg}(T)$ . This transformation allows us to convert the preliminary bound into a new, more defined upper bound via the divide-and-conquer method. Subsequently, we leverage the Arithmetic-Logarithmic-Geometric Mean inequality and other related inequality techniques to scale the new upper bound. Finally, we apply the Chernoff-Hoeffding inequality to two newly established upper bounds, thereby obtaining the explicit upper bound for  $\text{Reg}(T)$ .

*Remark 3:* Different from conventional methods, the upper bound in our method has a logarithmic function of channel gain rather than a linear one. The utilization of a logarithmic function complicates the derivation of the upper bound, rendering it insensitive to traditional methods. To this end, we adopt a series of inequality scaling, such as the Arithmetic-Logarithmic-Geometric Mean inequality, to convert the logarithmic function into a linear function. In this way, we can conveniently apply the Chernoff-Hoeffding inequality to calculate the explicit upper bound of  $\text{Reg}(T)$  with high accuracy.

*Remark 4:* Theorem 2 reveals that the second term of the upper bound for  $\text{Reg}(T)$  is  $O(\sqrt{\ln T/T})$ . This suggests that as  $T$  grows, the term  $\frac{\tau B_c P_{\max} U}{T \sigma_{\min}^2 \ln 2} F(T)$  decreases and finally tends towards zero. Consequently, with an increasing number of time slots, the upper bound for the time-averaged regret progressively reduces and ultimately converges to a constant  $\frac{B_1 U}{V}$ . Notably,  $\frac{B_1 U}{V}$  is relatively small when  $V$  is large, but becomes significant when  $V$  is small. In real-world scenarios, it is advisable to choose an appropriately large value for  $V$  to maintain a minimal ratio of  $\frac{B_1 U}{V}$ . This strategy is advantageous because a larger  $V$  typically maximizes the volume of data that can be offloaded.

**C. Online Data Offloading Scheme**

Here, we utilize the proposed solving framework in Section IV-B to address **P0** in a low computation cost. Specifically, we first introduce the following new *Lyapunov function*:

$$L_2(H_u(t)) = \frac{1}{2}Z_u(t)^2 + \frac{1}{2}Q_u(t)^2, \forall u, t, \quad (27)$$

where  $H_u(t) = (Z_u(t), Q_u(t))$  denotes the concatenated vector of the virtual queue and the actual queue. We further introduce the following *conditional Lyapunov drift* for any LEO satellite  $u$  in time slot  $t$  as follows:

$$\Delta_2(H_u(t)) = \mathbb{E}\{L_2(H_u(t+1)) - L_2(H_u(t)) | H_u(t)\}. \quad (28)$$

As such, the stability of  $Z_u(t)$  and  $Q_u(t)$  can be equivalently transformed into minimizing  $\Delta_2(H_u(t))$  [72].

Then, we use the DPP metric to equivalently transform **P0** into the following optimization problem:

$$\begin{aligned} \text{P8: } \min_{P(t), x(t)} \lim_{|\mathcal{T}| \rightarrow \infty} \frac{1}{|\mathcal{T}|} \sum_{\substack{t \in \mathcal{T} \\ u \in \mathcal{U}}} (\Delta_2(H_u(t)) - V \mathbb{E}\{b_u(t)\}) \\ \text{s.t. C2-C4.} \end{aligned}$$

We use (27) and (28) to yield:

$$\begin{aligned} \Delta_2(H_u(t)) - V \mathbb{E}\{b_u(t)\} &\leq B_2 - V \mathbb{E}\{b_u(t)\} \\ &+ \mathbb{E}\{Z_u(t)(e_u(t) - e_u^{\text{avg}}) + Q_u(t)(a_u(t) - b_u(t)) | H_u(t)\}, \end{aligned}$$

where  $B_2 = \frac{1}{2}(P_{\max}\tau - e_u^{\text{avg}})^2 + \frac{1}{2}(B_c \log_2(1 + \frac{\Psi_{\max}}{\sigma_g^2(t)} P_{\max})\tau + a(t))^2$ . Thus, we can obtain an upper bound problem of **P8**:

$$\begin{aligned} \text{P9: } \min_{P(t), x(t)} \lim_{|\mathcal{T}| \rightarrow \infty} \frac{1}{|\mathcal{T}|} \sum_{\substack{t \in \mathcal{T} \\ u \in \mathcal{U}}} \left( \mathbb{E}\{Z_u(t)(e_u(t) - e_u^{\text{avg}}) | H_u(t)\} \right. \\ \left. + \mathbb{E}\{Q_u(t)(a_u(t) - b_u(t)) | H_u(t)\} - V \mathbb{E}\{b_u(t)\} \right) + B_2 \\ \text{s.t. C2-C4.} \end{aligned}$$

Furthermore, we remove  $B_2$  from **P9** and then equivalently convert it into the following per-time-slot optimization subproblem for any LEO satellite  $u$  at any time slot  $t$ :

$$\begin{aligned} \text{P10: } \min_{P(t), x(t)} \sum_{g \in \mathcal{G}} x_{u,g}(t) \mathfrak{R}_{u,g}(t) \\ \text{s.t. C6-C8,} \end{aligned}$$

with  $\mathcal{R}_{u,g}(t) = (Z_u(t)P_{u,g}(t) - (Q_u(t) + V)R_{u,g}(t))d_{u,g}(t)$ . We further separate **P10** into the following two subproblems:

$$\mathbf{P11:} \min_{\mathbf{P}(t)} \mathcal{R}_{u,g}(t)$$

$$\text{s.t. C6,}$$

and

$$\mathbf{P12:} \min_{\mathbf{x}(t)} \sum_{g \in \mathcal{G}} x_{g,t} \mathcal{R}_{u,g}^*(t)$$

$$\text{s.t. C7-C8,}$$

where  $\mathcal{R}_{u,g}^*(t)$  is an optimal value to **P11**.

To solve **P11** for obtaining an optimal solution, we first need to obtain two extremal points of  $\mathcal{R}_{u,g}(t)$ , given by

$$P_{u,g,2}^0(t) = \frac{(Q_u(t) + V)B_c}{Z_u(t) \ln 2} - \frac{\sigma_g^2(t)}{\Psi_{u,g}(t)}, \quad (29)$$

and  $P_0^{u,g}(t)$ . If  $P_{u,g}(t) \in [P_{\min}, P_0^{u,g}(t)]$ , we can obtain the optimal solution for minimizing  $\mathcal{R}_{u,g}(t)$  as follows:

$$P_2^{u,g}(t) = \begin{cases} P_0^{u,g}(t) & Z_u(t) < \frac{\Psi_{u,g}(t)(Q_u(t)+V)B_c}{(\sigma_g^2(t)+\Psi_{u,g}(t)P_0^{u,g}(t)) \ln 2}, \\ P_{\min} & Z_u(t) > \frac{\Psi_{u,g}(t)(Q_u(t)+V)B_c}{(\sigma_g^2(t)+\Psi_{u,g}(t)P_{\min}) \ln 2}, \\ P_{u,g,2}^0(t) & \text{otherwise.} \end{cases} \quad (30)$$

Similar to (15) and (16), define the following two functions:

$$\Phi_{P_{u,g}(t)}^2 = \log_2 \left( \frac{2^{Z_u(t)P_{u,g}(t)}}{(1 + \frac{\Psi_{u,g}(t)}{\sigma_g^2(t)} P_{u,g}(t))(Q_u(t)+V)B_c} \right), \quad (31)$$

and

$$\varphi_{\eta_{u,g}(t)}^2 = \frac{\sigma_g^2(t)Z_u(t)\eta_{u,g}(t)}{\Psi_{u,g}(t)} \left( 2^{\frac{1}{B_c\eta_{u,g}(t)}} - 1 \right) - Q_u(t) - V. \quad (32)$$

The following theorem shows a closed-form optimal solution to **P11**.

*Theorem 3:* The optimal solution  $P_{u,g}^*(t)$  for **P11** is

$$P_{u,g}^*(t) = \begin{cases} P_{u,g,3}^*(t) & P_0^{u,g}(t) \in [P_{\min}, P_{\max}], \\ P_{u,g,4}^*(t) & P_0^{u,g}(t) \in (P_{\max}, \infty), \\ P_{\min} & P_0^{u,g}(t) \in (0, P_{\min}), \end{cases} \quad (33)$$

where  $P_{u,g,3}^*(t)$  and  $P_{u,g,4}^*(t)$  are respectively defined as

$$P_{u,g,3}^*(t) = \begin{cases} P_2^{u,g}(t) & \frac{\Phi_{P_2^{u,g}(t)}^2}{Q_u(t)} \leq \frac{\varphi_{\chi(P_0^{u,g}(t))}^2}{\tau}, \\ P_0^{u,g}(t) & \text{otherwise,} \end{cases} \quad (34)$$

and

$$P_{u,g,4}^*(t) = \begin{cases} P_{\max} & Z_u(t) < \frac{\Psi_{u,g}(t)(Q_u(t)+V)B_c}{(\sigma_g^2(t)+\Psi_{u,g}(t)P_{\max}) \ln 2}, \\ P_{\min} & Z_u(t) > \frac{\Psi_{u,g}(t)(Q_u(t)+V)B_c}{(\sigma_g^2(t)+\Psi_{u,g}(t)P_{\min}) \ln 2}, \\ P_{u,g,2}^0(t) & \text{otherwise.} \end{cases} \quad (35)$$

*Proof:* See Appendix C. ■

From Theorem 3, it is evident that the value of  $P_{u,g}^*(t)$  is related to both  $Q_u(t)$  and  $Z_u(t)$  queues. Typically, a longer length of  $Q_u(t)$  may yield a greater power allocation value, while a longer length of  $Z_u(t)$  may result in a smaller power allocation value. This outcome is due to our adoption of the DPP metric to stabilize both  $Q_u(t)$  and  $Z_u(t)$  queues.

*Remark 5:* Compared with **P1**, **P0** has one more constraint, i.e., C5. Intuitively, this suggests that the optimized objective value of **P0** should be less than that of **P1**. However, upon comparing the results from (20) and (35), it becomes clear that the optimal power allocation  $P_{u,g}^*(t)$  derived from Theorem 3 is actually larger than that derived from Theorem 1. This indicates that for problem **P0**, at each time slot, the optimal power allocation obtained in Theorem 3 is larger than that obtained in Theorem 1. Therefore, EODO-DQ can allocate more power at each time slot than EODO-SQ, thus obtaining larger data amount. This point is further verified by simulation results presented in Section V. Thus, Theorem 3 gives a conclusion opposite to the intuition that despite having an additional constraint, the optimized objective value of **P0** is larger than that of **P1**. The above discussion illustrates that stabilizing the system queues can enhance network performance of LSNs.

We can use the proposed UCB policy in Section IV-B to solve **P12** directly, since it has the same form as **P6**. We term the solution to **P8** as Energy-constrained Online Data Offloading with Double Queue (EODO-DQ) and summarize it as Algorithm 2. The following theorem indicates that EODO-DQ is also a performance-gap bounded algorithm.

*Theorem 4:* The theoretical upper bound for the time-averaged regret achieved by EODO-DQ is

$$\text{Reg}(T) \leq \frac{B_2 U}{V} + \frac{\tau B_c P_{\max} U}{T \sigma_{\min}^2 \ln 2} F(T). \quad (36)$$

*Proof:* See Appendix D. ■

*Remark 6:* It is easy to observe from Algorithm 1 and Algorithm 2 that the computational complexity of both Algorithms are dominated by calculating the optimal power allocation  $P_{u,g}^*(t)$  (Steps 6-21) and obtain  $\hat{x}_{u,g}^*(t)$  (Step 25). In each step, the complexity of calculating  $P_{u,g}^*(t)$  is  $O(1)$ . The complexity of obtaining  $\hat{x}_{u,g}^*(t)$  is  $O(G)$ . Therefore, the total computational complexity of both Algorithm 1 and Algorithm 2 are  $O(TUG)$ . The proposed EODO-SQ and EODO-DQ algorithms have a low computation cost, which is suitable to the online data offloading scenario in LSNs.

## V. NUMERICAL RESULTS

### A. Simulation Settings

We evaluate the proposed EODO-SQ and EODO-DQ algorithms through the following extensive simulations. The LSN simulation scenario comprises 12 LEO satellites and 100 GSs. The geographical coordinates of GSs are extracted from the Starlink GS map, and their details are listed in Table IV. The total duration is divided into 86400 time slots, each lasting 1 s, i.e.,  $\tau = 1$ . The data amount of each LEO satellite is generated by a Poisson distribution with  $\lambda = 25\text{Mb}$ , where  $\lambda$  denotes the average data arrival rate of LSN. The remaining parameters

**Algorithm 2:** Proposed EODO-DQ Algorithm.

---

**Input:**  $\mathcal{G}_u(t)$ ,  $P_{\min}$ ,  $P_{\max}$ ,  $\tau$ ,  $B_c$ ,  $e_u^{\text{avg}}$ ,  $V$ .

- 1: **Initialization:**  $\bar{\Psi}_{u,g}(0) = 0$ ,  $n_{u,g}(0) = 0$ ,  $Q_u(0) = 0$ ,  $Z_u(0) = 0$ .
- 2: **for**  $t = 1, \dots, T$  **do**
- 3:   **for**  $u = 1, \dots, U$  **do**
- 4:     **for**  $g = 1, \dots, G$  **do**
- 5:       **if**  $g \in \mathcal{G}_u(t)$  **then**
- 6:          Use (21) to compute  $\hat{\Psi}_{u,g}(t)$ .
- 7:          Use (12) to obtain  $P_{0,g}^{u,g}(t)$ .
- 8:          Use (29) to obtain  $P_{2,g}^{u,g}(t)$ .
- 9:          Use (30) to compute  $P_{2,g}^{u,g}(t)$ .
- 10:         Use (31) to obtain  $\Phi_{P_{2,g}^{u,g}(t)}^2$ .
- 11:         Use (32) to obtain  $\varphi_{\chi(P_{0,g}^{u,g}(t))}^2$ .
- 12:         **if**  $P_{0,g}^{u,g}(t) \in [P_{\min}, P_{\max}]$  **then**
- 13:           **if**  $\Phi_{P_{2,g}^{u,g}(t)}^2 \leq \varphi_{\chi(P_{0,g}^{u,g}(t))}^2 Q_u(t)$  **then**
- 14:              $\hat{P}_{u,g}^*(t) = P_{2,g}^{u,g}(t)$ .
- 15:           **else**
- 16:              $\hat{P}_{u,g}^*(t) = P_{0,g}^{u,g}(t)$ .
- 17:           **end if**
- 18:         **else if**  $P_{0,g}^{u,g}(t) \in (P_{\max}, \infty)$  **then**
- 19:           Use (35) to obtain  $\hat{P}_{u,g}^*(t)$ .
- 20:         **else**
- 21:            $\hat{P}_{u,g}^*(t) = P_{\min}$ .
- 22:         **end if**
- 23:       **end if**
- 24:     **end for**
- 25:     Use (24) to obtain  $\hat{x}_{u,g}^*(t)$ .
- 26:     Use (5) and (11) to update  $Q_u(t)$  and  $Z_u(t)$ .
- 27:     Use (22) and (23) to update  $\bar{\Psi}_{u,g}(t)$  and  $n_{u,g}(t)$ .
- 28:   **end for**
- 29: **end for**

**Output:**  $\hat{P}^*(t) = \{\hat{P}_{u,g}^*(t)\}$ ,  $\hat{x}^*(t) = \{\hat{x}_{u,g}^*(t)\}$ .

---

are set as follows:  $\xi_{\text{CL}} = 17$  dB,  $\xi_e = 13.6$  dB,  $\xi_{\text{rain}} = 0$  dB,  $B_c = 3.5$  GHz,  $f_c = 20$  GHz,  $h_g^{\text{tr}}(t) = 43.2$  dB,  $h_g^{\text{re}}(t) = 39.7$  dB,  $e_u^{\text{avg}} = 2$  J,  $V = 2.0 \times 10^6$ ,  $P_{\min} = 0$  W,  $P_{\max} = 4$  W, and the mean of  $\xi_{\text{SF}}$  is 1.9 dB.

**B. Design Elements of EODO-SQ and EODO-DQ**

To show the convergence evolution of EODO-SQ and EODO-DQ with different values of  $\lambda$ , Fig. 3(a) and (b) plot the total offloaded data amount and the time-averaged energy consumption versus the value of parameter  $V$ , respectively. From Fig. 3(a), we observe that the total offloaded data amount under different values of  $\lambda$  first increases and then inevitably converges to a maximum value as the value of  $V$  grows. This indicates that we can set  $V$  large enough to guarantee the convergence of EODO-SQ and EODO-DQ. From Fig. 3(b), we observe that the time-averaged energy consumption under different values of  $\lambda$  increases as the value of  $V$  grows. This phenomenon illustrates that a larger value of  $V$  would increase energy consumption. The above observations from Fig. 3(a) and (b) indicate that setting parameter  $V$  enables the balance of maximizing the total offloaded

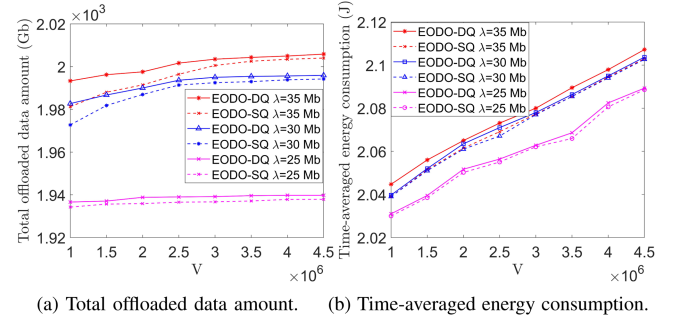
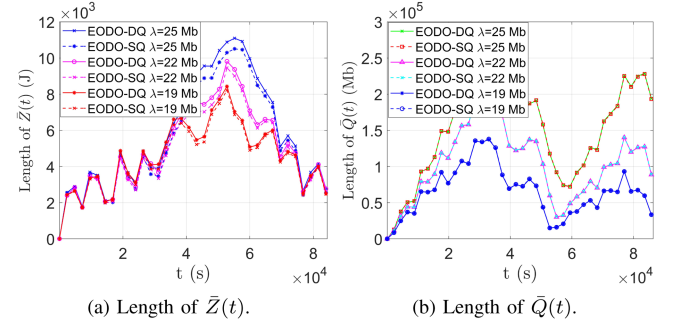
Fig. 3. Verification convergence versus  $V$ .

Fig. 4. Average lengths of two queue types versus time slots.

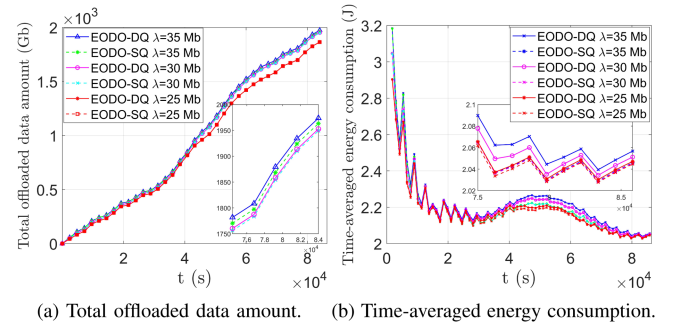


Fig. 5. Various network metrics versus time slots.

data amount and minimizing the energy consumption. Thus, setting a suitable value of  $V$  for EODO-SQ and EODO-DQ enables the large total offloaded data amount with less energy consumption.

Fig. 4(a) and (b) respectively show the average length of virtual queue  $\bar{Z}(t) = \frac{1}{U} \sum_{u \in U} Z_u(t)$  and the average length of actual queue  $\bar{Q}(t) = \frac{1}{U} \sum_{u \in U} Q_u(t)$  versus time slots for the different settings of  $\lambda$ . From Fig. 4(a) and (b), it is observed that both EODO-SQ and EODO-DQ stabilize the queues of  $\bar{Z}(t)$  and  $\bar{Q}(t)$  for different values of  $\lambda$ . This observation reflects that both EODO-SQ and EODO-DQ can guarantee C1 and C5 in P0. Furthermore, Fig. 4(b) shows that the length of  $\bar{Q}(t)$  attained by both EODO-SQ and EODO-DQ with a larger value of  $\lambda$  is greater than that achieved with a smaller value of  $\lambda$ . This is attributed to the fact that a larger value of  $\lambda$  implies more data being fed to the LSN, which in turn increases the length of  $\bar{Q}(t)$ .

TABLE IV  
THE GEOGRAPHICAL COORDINATES OF GSS

|                           |                           |                           |                           |
|---------------------------|---------------------------|---------------------------|---------------------------|
| GS1 (5.33°, 163.339°)     | GS2 (−10.498°, −53.345°)  | GS3 (−35.848°, −82.585°)  | GS4 (−4.389°, −73.871°)   |
| GS5 (20.247°, −21.597°)   | GS6 (30.019°, −20.662°)   | GS7 (9.567°, −10.322°)    | GS8 (0.656°, −0.612°)     |
| GS9 (0.848°, 176.976°)    | GS10 (47.896°, 167.081°)  | GS11 (−23.069°, −38.287°) | GS12 (5.402°, −28.694°)   |
| GS13 (−5.142°, −116.693°) | GS14 (27.064°, 30.989°)   | GS15 (32.065°, 43.866°)   | GS16 (11.9°, 53.116°)     |
| GS17 (10.488°, 48.553°)   | GS18 (1.464°, 9.258°)     | GS19 (−8.21°, 7.325°)     | GS20 (−19.62°, 16.993°)   |
| GS21 (−37.977°, 28.594°)  | GS22 (−36.296°, 42.648°)  | GS23 (−14.266°, 95.452°)  | GS24 (6.349°, 78.819°)    |
| GS25 (4.487°, 51.568°)    | GS26 (25.32°, 57.841°)    | GS27 (23.77°, 36.75°)     | GS28 (−1.35°, 7.04°)      |
| GS29 (4.833°, 25.67°)     | GS30 (−25.241°, 32.415°)  | GS31 (−25.921°, −2.402°)  | GS32 (51.706°, −0.601°)   |
| GS33 (38.608°, 15.659°)   | GS34 (41.433°, 23.842°)   | GS35 (54.879°, 29.509°)   | GS36 (40.789°, −7.478°)   |
| GS37 (40.265°, 4.687°)    | GS38 (52.291°, 9.039°)    | GS39 (45.475°, 18.942°)   | GS40 (69.662°, 23.384°)   |
| GS41 (42.672°, 140.614°)  | GS42 (36.387°, 141.258°)  | GS43 (43.173°, 131.556°)  | GS44 (34.217°, 178.468°)  |
| GS45 (−18.129°, 174.301°) | GS46 (−35.794°, 146.422°) | GS47 (−20.0°, 122.511°)   | GS48 (−3.977°, 168.381°)  |
| GS49 (−46.529°, 141.441°) | GS50 (−31.998°, 107.097°) | (−6.306°, 113.925°)       | (−2.219°, 118.283°)       |
| GS53 (−31.496°, 147.091°) | GS54 (−38.097°, 153.055°) | GS55 (−27.068°, 123.922°) | GS56 (10.296°, 101.367°)  |
| GS57 (3.02°, −109.926°)   | GS58 (22.913°, −147.5°)   | GS59 (64.805°, −95.815°)  | GS60 (40.334°, −102.89°)  |
| GS61 (30.194°, −99.811°)  | GS62 (48.66°, −165.428°)  | GS63 (64.504°, −63.613°)  | GS64 (44.465°, −66.192°)  |
| GS65 (18.431°, −52.775°)  | GS66 (47.561°, −59.538°)  | GS67 (13.063°, −159.378°) | GS68 (21.999°, −92.933°)  |
| GS69 (18.041°, 144.859°)  | GS70 (13.616°, −95.3°)    | GS71 (32.614°, −111.859°) | GS72 (33.421°, −85.877°)  |
| GS73 (33.594°, −122.449°) | GS74 (48.774°, −112.225°) | GS75 (44.17°, −88.009°)   | GS76 (43.151°, −104.617°) |
| GS77 (39.758°, −119.91°)  | GS78 (36.859°, −100.274°) | GS79 (20.573°, −77.465°)  | GS80 (39.016°, −73.425°)  |
| GS81 (45.567°, −43.833°)  | GS82 (−16.684°, −56.527°) | GS83 (−29.765°, −59.998°) | GS84 (−2.927°, −70.883°)  |
| GS85 (−33.393°, −70.85°)  | GS86 (−52.94°, −68.908°)  | GS87 (12.098°, −37.147°)  | GS88 (−11.081°, −46.863°) |
| GS89 (−23.466°, −38.462°) | GS90 (−3.735°, −77.042°)  | GS91 (−12.148°, −65.202°) | GS92 (−40.389°, −65.442°) |
| GS93 (−24.805°, −60.068°) | GS94 (−34.92°, −75.498°)  | GS95 (6.106°, −166.505°)  | GS96 (53.86°, 166.918°)   |
| GS97 (−0.529°, −80.667°)  | GS98 (28.543°, −130.187°) | GS99 (56.466°, −68.396°)  | GS100 (58.11°, 163.017°)  |

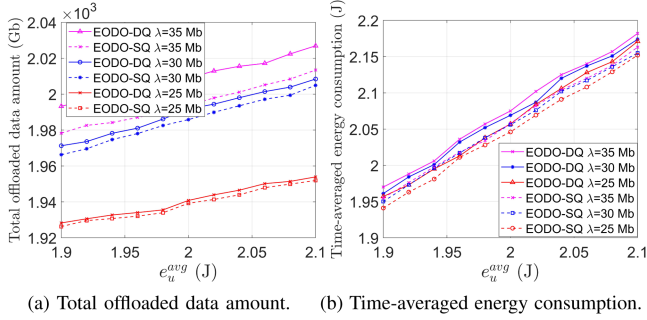


Fig. 6. Various network metrics versus  $e_u^{\text{avg}}$ .

In Fig. 5(a) and (b), we plot the total offloaded data amount and the time-averaged energy consumption versus time slots for the different settings of  $\lambda$ , respectively. We observe from Fig. 5(a) that the total offloaded data amount of both EODO-SQ and EODO-DQ increase with the growing number of time slots. Meanwhile, Fig. 5(b) shows that the time-averaged energy consumption of both EODO-SQ and EODO-DQ decreases and then converges to  $e_u^{\text{avg}}$  as the number of time slots increases. The above observations illustrate that both EODO-SQ and EODO-DQ can efficiently offload more data amount while guaranteeing C1. From Fig. 5(a) and (b), it can be seen that both EODO-DQ and EODO-SQ with a larger value of  $\lambda$  can offload larger data amount and consume more energy than a smaller one. This is accounted that the more data amount the larger energy consumption.

Fig. 6(a) plots the total offloaded data amount against  $e_u^{\text{avg}}$  for different settings of  $\lambda$ . We can see from Fig. 6(a) that the total offloaded data amount increases with the increase of  $e_u^{\text{avg}}$ . This is because a large value of  $e_u^{\text{avg}}$  leads to a large range of power allocation for both EODO-SQ and EODO-DQ in each time slot, thus increasing the total offloaded data amount of LSN. Fig. 6(a)

further shows that EODO-DQ outperform EODO-SQ in terms of total offloaded data amount versus  $e_u^{\text{avg}}$  under different values of  $\lambda$ . The reason behind is that EODO-DQ is capable of allocating a greater optimal power than that of EODO-SQ in each time slot to guarantee the mean rate queue stability, while satisfying the constraint of long-term time-averaged energy consumption. This phenomenon also illustrates that enhancing the stability of system queues can improve the efficiency of data offloading in LSNs.

Fig. 6(b) plots the time-averaged energy consumption against  $e_u^{\text{avg}}$ . We can see from Fig. 6(b) that the time-averaged energy consumption increases with the increase of  $e_u^{\text{avg}}$ . It is also noted that when  $\lambda$  is fixed, the data offloaded by EODO-DQ is higher than that offloaded by EODO-SQ. This because  $P_{u,g,2}^0(t)$  is greater than  $P_{u,g,1}^0(t)$ , which results in EODO-DQ consuming more energy than EODO-SQ.

### C. Performance Comparisons

To further evaluate the performance of EODO-SQ and EODO-DQ, we adopt five baseline algorithms involving the EODO-DQ with Random GS selection (EODO-RND) algorithm, the MAB with Random power allocation (MAB-RND) algorithm, the Random (RND) algorithm, the Optimal Power allocation and Task scheduling (OPT) algorithm, and the DAC [66] algorithm. The details of these baseline algorithm are listed as follows:

- The EODO-RND algorithm: It uses EODO-DQ to optimize power control but randomly selects one available GS in each time slot.
- The MAB-RND algorithm: It utilizes the UCB policy to solve **P12** but optimizes the power allocation in a random manner within  $[P_{\min}, P_{\max}]$ .

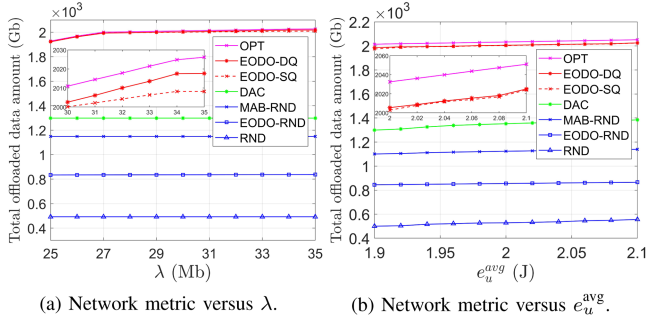


Fig. 7. Comparisons with alternatives.

- The RND algorithm: In this scheme, each GS is randomly selected and the power allocation of each time slot is randomly generated within  $[P_{\min}, P_{\max}]$ .
- The OPT algorithm leverages the **P11** solution in Section IV-C to obtain the optimal power allocation and selects one GS associated with the best objective value of **P12** in each time slot with the known channel gain of  $\Psi_g(t), \forall g$ .
- The DAC algorithm: It is an efficient deep reinforcement learning based algorithm consisting of the two neural networks of the actor network and the critic network. The actor network is responsible for making the joint decisions on the GS selection and the power allocation. Then, the critic network evaluates the generated decisions in terms of the offloaded data amount and energy consumption. The interaction of the two networks as the time slot evolving promotes the decision efficiency of the actor network while enhancing the evaluation accuracy of the critic network.

In Fig. 7(a) and (b), we compare EODO-SQ and EODO-DQ with the five baseline algorithms in terms of the total offloaded data amount versus the values of  $\lambda$  and  $e_u^{\text{avg}}$ , respectively. From Fig. 7(a) and (b), we observe that both EODO-SQ and EODO-DQ offload more task data than the schemes of EODO-RND, MAB-RND, and RND as the values of both  $\lambda$  and  $e_u^{\text{avg}}$  increase. This is because both EODO-SQ and EODO-DQ jointly optimize the variables of the power allocation and task scheduling to improve the total offloaded data amount of the LSN, while the alternatives only optimize at most one optimization variable.

Through the joint optimization of power allocation and task scheduling variables by EODO-DQ and EODO-SQ, we can see from Fig. 7(a) and (b) that both algorithms can approach the performance of OPT scheme in terms of the total offloaded data amount as  $\lambda$  and  $e_u^{\text{avg}}$  increase. This indicates that our proposed algorithms, i.e., EODO-SQ and EODO-DQ, are very effective for improving the data offload efficiency of LSNs. We can also observe from Fig. 7(a) that EODO-DQ performs slightly better than EODO-SQ when  $\lambda > 30\text{Mb}$ . This is because when  $\lambda$  is large, EODO-DQ increases the power to ensure the stability of  $Q_u(t)$ , which achieves more data offload than EODO-SQ.

Fig. 7(a) and (b) show that the total offloaded data amount of DAC, EODO-RND, MAB-RND, and RND remains constant, even with an increase in  $\lambda$  and  $e_u^{\text{avg}}$ . The reasons are analyzed as follows. The DAC algorithm fails to adequately account for the

dynamic nature of SGLs, and therefore, it cannot guarantee the stability of queues  $Q_u(t)$  and  $Z_u(t)$ . As such, its total offloaded data amount is less than that of our proposed algorithms when  $\lambda$  increases. Although EODO-RND is capable of stabilizing queue  $Z_u(t)$ , the inherent unpredictability of GS selection results in the degradation of its data offloading efficiency. This subsequently affects the stabilization of queue  $Q_u(t)$ , causing its total offloaded data amount to be lower than that of both EODO-DQ and EODO-SQ. In particular, the schemes of MAB-RND and RND do not succeed in stabilizing any queue, thereby demonstrating substantially inferior data offloading performance.

## VI. CONCLUSION

In this paper, we have studied the energy-constrained online data offloading in LSNs via joint optimization of power allocation and task scheduling, accounting for the stochastic nature of space data arrivals and the dynamic SGLs. We first explore the studied problem without the mean rate queue stability constraint, and then decompose it into a sequence of per-time-slot subproblems. After exploiting the distinct features of each subproblem, an efficient two-layer optimization method is presented. On this basis, we further extend this solution to solve the original problem. Furthermore, through theoretical analyses, we show that the performance gap between our proposed algorithms and the theoretical optimal solution is bounded. The simulation results show that the proposed algorithms significantly outperform existing state-of-the-art solutions, while highlighting the relationship between the total offloaded data volume and energy consumption in LSNs and closely approximating the optimal strategy. The simulation results also demonstrate that both the enhancement of system queue stability and the joint online optimization of power allocation and task scheduling can improve the data efficiency of LSNs.

## APPENDIX A PROOF OF THEOREM 1

The definition of  $d_{u,g}(t)$  in (9) indicates that it is a piecewise function. As such, we discuss two cases of this piecewise function. We note that the partition point of  $d_{u,g}(t)$  can be calculated as (12). Since  $\frac{Q_u(t)}{\tau B_c} \geq 0$ , we have three cases for the range of  $P_0^{u,g}(t)$ :  $P_0^{u,g}(t) \in [P_{\min}, P_{\max}]$ ,  $P_0^{u,g}(t) \in (P_{\max}, \infty)$ , and  $P_0^{u,g}(t) \in (0, P_{\min})$ .

Case 1: Since  $P_0^{u,g}(t) \in [P_{\min}, P_{\max}]$ , we recast (9) as

$$d_{u,g}(t) = \begin{cases} \tau & P_{u,g}(t) \in [P_{\min}, P_0^{u,g}(t)], \\ \frac{Q_u(t)}{R_{u,g}(t)} & P_{u,g}(t) \in (P_0^{u,g}(t), P_{\max}]. \end{cases}$$

When  $P_{u,g}(t) \in [P_{\min}, P_0^{u,g}(t)]$ , we have  $d_{u,g}(t) = \tau$  and rewrite **P5** as the following form:

$$\begin{aligned} \textbf{P13: } \min_{P_{u,g}(t)} & (Z_u(t)P_{u,g}(t) - VR_{u,g}(t))\tau \\ \text{s.t. } & P_{\min} \leq P_{u,g}(t) \leq P_0^{u,g}(t). \end{aligned}$$

Furthermore, we substitute (1) and (2) into **P13** to yield:  $(Z_u(t)P_{u,g}(t) - VB_c \log_2(1 + \frac{\Psi_{u,g}(t)}{\sigma_g^2(t)} P_{u,g}(t)))\tau$ . For the convenience of analysing, we define  $\Phi_{P_{u,g}(t)}^1$  as shown in (15). Then, we deduce the first-order and second-order derivations of  $\Phi_{P_{u,g}(t)}^1$  as follows:

$$\nabla \Phi_{P_{u,g}(t)}^1 = Z_u(t) - \frac{VB_c \Psi_{u,g}(t)}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{u,g}(t)) \ln 2}, \quad (37)$$

and

$$\nabla^2 \Phi_{P_{u,g}(t)}^1 = \frac{VB_c (\Psi_{u,g}(t))^2}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{u,g}(t))^2 \ln 2}. \quad (38)$$

From (38), we see that  $\nabla^2 \Phi_{P_{u,g}(t)}^1 \geq 0$ , indicating that **P13** is convex. Thus, we use  $\nabla \Phi_{P_{u,g}(t)}^1 = 0$  to compute an extremal point to **P13**, given by (13). Let  $P_{u,g}^{1,g}(t)$  be the optimal solution to **P13** and consider the following three cases: If  $P_{\min} \leq P_{u,g,1}^0(t) \leq P_0^{u,g}(t)$ , we use (13) to yield:

$$\frac{VB_c}{\left(\frac{\sigma_g^2(t)}{\Psi_{u,g}(t)} + P_0^{u,g}(t)\right) \ln 2} \leq Z_u(t) \leq \frac{VB_c}{\left(\frac{\sigma_g^2(t)}{\Psi_{u,g}(t)} + P_{\min}\right) \ln 2}.$$

Thus, we have  $P_{u,g}^{1,g}(t) = P_{u,g,1}^0(t)$ . If  $P_{u,g,1}^0(t) > P_0^{u,g}(t)$ , we have  $Z_u(t) < \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_0^{u,g}(t)) \ln 2}$ . When  $P_{\min} \leq P_{u,g}(t) \leq P_0^{u,g}(t)$ , we have  $\nabla \Phi_{P_{u,g}(t)}^1 < 0$ . Thus, we obtain  $P_{u,g}^{1,g}(t) = P_0^{u,g}(t)$ . If  $P_{u,g,1}^0(t) < P_{\min}$ , we have  $Z_u(t) > \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\min}) \ln 2}$ . When  $P_{\min} \leq P_{u,g}(t) \leq P_0^{u,g}(t)$ , we have  $\nabla \Phi_{P_{u,g}(t)}^1 > 0$ . Thus, we have  $P_{u,g}^{1,g}(t) = P_{\min}$ . In brief,  $P_{u,g}^{1,g}(t)$  is given by the form of (14).

When  $P_{u,g}(t) \in (P_0^{u,g}(t), P_{\max}]$ , we have  $d_{u,g}(t) = \frac{Q_u(t)}{R_{u,g}(t)}$  and rewrite **P5** as the following problem:

$$\begin{aligned} \text{P14: } \min_{P_{u,g}(t)} & \left( Z_u(t) \frac{P_{u,g}(t)}{R_{u,g}(t)} - V \right) Q_u(t) \\ \text{s.t. } & P_0^{u,g}(t) \leq P_{u,g}(t) \leq P_{\max}. \end{aligned}$$

For facilitating problem solving, we introduce a new variable  $\eta_{u,g}(t) = \chi(P_{u,g}(t))$  to replace  $P_{u,g}(t)$  and  $R_{u,g}(t)$ , where  $\chi(\cdot)$  is defined as (17). As such, we can easily derive:

$$P_{u,g}(t) = \frac{\sigma_g^2(t)}{\Psi_{u,g}(t)} \left( 2^{\frac{1}{B_c \eta_{u,g}(t)}} - 1 \right), \quad (39)$$

and

$$R_{u,g}(t) = \frac{1}{\eta_{u,g}(t)}. \quad (40)$$

On the basis of  $\eta_{u,g}(t)$  and  $\chi(\cdot)$ , we further use (39) and (40) to recast **P14** as the following form:

$$\begin{aligned} \text{P15: } \min_{\eta_{u,g}(t)} & \varphi_{\eta_{u,g}(t)}^1 Q_u(t) \\ \text{s.t. C9: } & \chi(P_{\max}) \leq \eta_{u,g}(t) \leq \chi(P_0^{u,g}(t)), \end{aligned}$$

where  $\varphi_{\eta_{u,g}(t)}^1$  is defined in (16). We further derive the first-order and second-order derivation of  $\varphi_{\eta_{u,g}(t)}^1$ :

$$\nabla \varphi_{\eta_{u,g}(t)}^1 = \frac{\sigma_g^2(t) Z_u(t) \left( 2^{\frac{1}{\eta_{u,g}(t) B_c}} \left( 1 - \frac{\ln 2}{\eta_{u,g}(t) B_c} \right) - 1 \right)}{\Psi_{u,g}(t)}, \quad (41)$$

and

$$\nabla^2 \varphi_{\eta_{u,g}(t)}^1 = \frac{\sigma_g^2(t) Z_u(t) (\ln 2)^2}{\Psi_{u,g}(t) (\eta_{u,g}(t))^3 (B_c)^2} 2^{\frac{1}{\eta_{u,g}(t) B_c}}. \quad (42)$$

From (42), we have  $\nabla^2 \varphi_{\eta_{u,g}(t)}^1 > 0$ , indicating that  $\nabla \varphi_{\eta_{u,g}(t)}^1$  monotonously increases as  $\eta_{u,g}(t)$  increasing. Note that the upper bound of  $\nabla \varphi_{\eta_{u,g}(t)}^1$  is zero, i.e.,  $\lim_{\eta_{u,g}(t) \rightarrow +\infty} \nabla \varphi_{\eta_{u,g}(t)}^1 = 0$ . Thus, when  $\chi(P_{\max}) \leq \eta_{u,g}(t) \leq \chi(P_0^{u,g}(t))$ , we can easily obtain  $\nabla \varphi_{\eta_{u,g}(t)}^1 < 0$ , revealing that  $\varphi_{\eta_{u,g}(t)}^1$  monotonously decreases as  $\eta_{u,g}(t)$  increasing. Thus, the optimal solution to **P15** is  $\chi(P_0^{u,g}(t))$ , further obtaining  $P_{u,g}^{2,g}(t) = P_0^{u,g}(t)$  to **P14**. In brief, the optimal power to **P13** is given by (19).

*Case 2:* Since  $P_0^{u,g}(t) \in (P_{\max}, \infty)$ , we have  $d_{u,g}(t) = \tau$ ,  $P_{u,g}(t) \in [P_{\min}, P_{\max}]$  and recast **P5** as the following form:

$$\begin{aligned} \text{P16: } \min_{P_{u,g}(t)} & (Z_u(t) P_{u,g}(t) - V R_{u,g}(t)) \tau \\ \text{s.t. C6.} \end{aligned}$$

Since **P16** is the same form as **P13**,  $P_{u,g,1}^0(t)$  is also an extremal point to **P16**. Thus, consider the following three cases: If  $P_{\min} \leq P_{u,g,1}^0(t) \leq P_{\max}$ , we use (13) to yield:

$$\frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\max}) \ln 2} \leq Z_u(t) \leq \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\min}) \ln 2}.$$

Then, the optimal solution to **P16** is  $P_{u,g,1}^0(t)$ . If  $P_{u,g,1}^0(t) > P_{\max}$ , we have  $Z_u(t) < \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\max}) \ln 2}$ . Then, the optimal solution to **P16** is  $P_{\max}$ . If  $P_{u,g,1}^0(t) < P_{\min}$ , we have  $Z_u(t) > \frac{\Psi_{u,g}(t)VB_c}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{\min}) \ln 2}$ . Thus, the optimal solution to **P16** is  $P_{\min}$ . In brief, the optimal solution to **P16** is given by (20).

*Case 3:* Since  $P_0^{u,g}(t) \in (0, P_{\min})$ , we have  $d_{u,g}(t) = \frac{Q_u(t)}{R_{u,g}(t)}$ ,  $P_{u,g}(t) \in [P_{\min}, P_{\max}]$  and recast **P5** as follows:

$$\begin{aligned} \text{P17: } \min_{P_{u,g}(t)} & \left( Z_u(t) \frac{P_{u,g}(t)}{R_{u,g}(t)} - V \right) Q_u(t) \\ \text{s.t. C6.} \end{aligned}$$

Next, we further transform **P17** into the following problem:

$$\begin{aligned} \text{P18: } \min_{\eta_{u,g}(t)} & \varphi_{\eta_{u,g}(t)}^1 Q_u(t) \\ \text{s.t. C10: } & \chi(P_{\max}) \leq \eta_{u,g}(t) \leq \chi(P_{\min}). \end{aligned}$$

Since **P18** has the same form of **P15**, we can obtain that  $\nabla^2 \varphi_{\eta_{u,g}(t)}^1 > 0$  and the upper bound of  $\nabla \varphi_{\eta_{u,g}(t)}^1$  is zero. Therefore, we have  $\nabla \varphi_{\eta_{u,g}(t)}^1 < 0$ , which indicates that  $\varphi_{\eta_{u,g}(t)}^1$  monotonously decreases as  $\eta_{u,g}(t)$  increasing. Hence, the optimal solution to **P18** is  $\chi(P_{\min})$ , and the optimal power to **P17** is obtained by

$$P_{u,g}^*(t) = P_{\min}. \quad (43)$$

Combining (19), (20), and (43), we can attain (18).

#### APPENDIX B PROOF OF THEOREM 2

We first define the per-time-slot regret as

$$\Delta \text{Reg}_u(t) = R_u^*(t)d_u^*(t) - \mathbb{E}\{b_u(t)\}. \quad (44)$$

Then, we use the DPP metric to calculate the drift-plus-regret as follows:

$$\begin{aligned} & \Delta_1(Z_u(t)) + V\mathbb{E}\{\Delta \text{Reg}_u(t)\} \\ & \leq B_1 + \mathbb{E}\{Z_u(t)(e_u(t) - e_u^{\text{avg}})|Z_u(t)\} + \Omega_u^1(t), \end{aligned} \quad (45)$$

where

$$\Omega_u^1(t) = V(R_u^*(t)d_u^*(t) - \mathbb{E}\{b_u(t)\}). \quad (46)$$

By taking expectation of both sides in inequality (45) and summarizing it over time slots, we have

$$\begin{aligned} & \mathbb{E}\{\Delta_1(Z_u(T))\} - \mathbb{E}\{\Delta_1(Z_u(1))\} + V \sum_{t=1}^T \mathbb{E}\{\Delta \text{Reg}_u(t)\} \\ & \leq TB_1 + \sum_{t=1}^T \mathbb{E}\{\Omega_u^1(t)\}. \end{aligned} \quad (47)$$

We divide both sides of (47) by  $TV$  to obtain

$$\mathbb{E}\left\{\frac{1}{T} \sum_{t=1}^T \Delta \text{Reg}_u(t)\right\} \leq \frac{B_1}{V} + \frac{1}{TV} \sum_{t=1}^T \mathbb{E}\{\Omega_u^1(t)\}. \quad (48)$$

Furthermore, we substitute (25) and (44) into (48) to yield:

$$\text{Reg}(T) \leq \frac{B_1}{V}U + \frac{1}{TV} \sum_{u \in \mathcal{U}} \sum_{t=1}^T \mathbb{E}\{\Omega_u^1(t)\}. \quad (49)$$

*a) Upper bound of  $\Omega_u^1(t)$ :* We decouple **P7** into a set of per-time-slot regret minimization subproblems for each LEO satellite, each given by

$$\begin{aligned} \text{P19: } & \min_{P_{u,g}(t), x_{u,g}(t)} R_u^*(t)d_u^*(t) - R_u(t)d_u(t) \\ & \text{s.t. C2-C4.} \end{aligned}$$

We let  $\{P_{u,g}'(t), x_{u,g}'(t)\}$  denote an optimal solution to **P19**. Since  $\{P_{u,g}^*(t), x_{u,g}^*(t)\}$  is a feasible solution to **P19**, we have

$$R_u^*(t)d_u^*(t) \leq R_u'(t)d_u'(t). \quad (50)$$

We combine (46) and (50) to obtain:

$$\Omega_u^1(t) \leq V(R_u'(t)d_u'(t) - \mathbb{E}\{R_u(t)d_u(t)\}). \quad (51)$$

Theorem 1 indicates that the inequality always holds:

$$\mathbb{E}\{\mathcal{R}(\hat{\Psi}_u(t), \hat{P}_u^*(t))\hat{d}_u(t)\} \geq \mathcal{R}(\hat{\Psi}_u(t), P_u'(t))d_u'(t), \quad (52)$$

where  $\mathcal{R}(\Psi_u(t), P_u(t)) = B_c \log_2(1 + \frac{\Psi_u(t)}{\sigma_g^2(t)} P_u(t))$ ,  $\hat{\Psi}_u(t) = \sum_{g \in \mathcal{G}_u(t)} \hat{x}_{u,g}^*(t)\hat{\Psi}_{u,g}(t)$ ,  $\hat{P}_u^*(t) = \sum_{g \in \mathcal{G}_u(t)} \hat{x}_{u,g}^*(t)\hat{P}_{u,g}^*(t)$ , and  $\hat{d}_u(t) = \sum_{g \in \mathcal{G}_u(t)} \hat{x}_{u,g}^*(t)\hat{d}_{u,g}(t)$  denotes the estimated

transmission time. Furthermore, we combine (51) and (52) to obtain:

$$\begin{aligned} \Omega_u^1(t) & \leq V(R_u'(t)d_u'(t) - \mathbb{E}\{R_u(t)d_u(t)\}) \\ & \quad + V(\mathbb{E}\{\mathcal{R}(\hat{\Psi}_u(t), \hat{P}_u^*(t))\hat{d}_u(t)\} \\ & \quad - \mathcal{R}(\hat{\Psi}_u(t), P_u'(t))d_u'(t)) \\ & = V \sum_{g \in \mathcal{G}_u(t)} \mathcal{R}(\Psi_{u,g}(t), P_{u,g}'(t))d_{u,g}'(t)x_{u,g}'(t) \\ & \quad - V \sum_{g \in \mathcal{G}_u(t)} \mathcal{R}(\Psi_{u,g}(t), \hat{P}_{u,g}^*(t))\hat{d}_{u,g}(t)\hat{x}_{u,g}^*(t) \\ & \quad + V \sum_{g \in \mathcal{G}_u(t)} \mathcal{R}(\hat{\Psi}_{u,g}(t), \hat{P}_{u,g}^*(t))\hat{d}_{u,g}(t)\hat{x}_{u,g}^*(t) \\ & \quad - V \sum_{g \in \mathcal{G}_u(t)} \mathcal{R}(\hat{\Psi}_{u,g}(t), P_{u,g}'(t))d_{u,g}'(t)x_{u,g}'(t) \\ & = V(\Omega_u^2(t) + \Omega_u^3(t)). \end{aligned} \quad (53)$$

In (53),  $\Omega_u^2(t)$  and  $\Omega_u^3(t)$  is defined as follows:

$$\Omega_u^2(t) = \sum_{g \in \mathcal{G}_u(t)} \kappa_{\hat{\Psi}_{u,g}(t), \Psi_{u,g}(t)}(\hat{P}_{u,g}^*(t))\hat{d}_{u,g}(t)\hat{x}_{u,g}^*(t), \quad (54)$$

and

$$\Omega_u^3(t) = \sum_{g \in \mathcal{G}_u(t)} \kappa_{\Psi_{u,g}(t), \hat{\Psi}_{u,g}(t)}(P_{u,g}'(t))d_{u,g}'(t)x_{u,g}'(t), \quad (55)$$

where  $\kappa_{\hat{\Psi}_{u,g}(t), \Psi_{u,g}(t)}(P_{u,g}'(t)) = B_c \log_2((\hat{\Psi}_{u,g}(t)P_{u,g}'(t) + \sigma_g^2(t))/(\Psi_{u,g}(t)P_{u,g}'(t) + \sigma_g^2(t)))$ .

*b) Upper bound of  $\Omega_u^2(t)$ :* We define event  $J_{u,g}(t) = \{\Psi_{u,g}(t) < \hat{\Psi}_{u,g}(t)\}$  for each  $g \in \mathcal{G}_u(t)$ , and recast (54) as the following form:

$$\Omega_u^2(t) = \sum_{g \in \mathcal{G}_u(t)} \kappa_{\hat{\Psi}_{u,g}(t), \Psi_{u,g}(t)}(\hat{P}_{u,g}^*(t))\varrho_{J_{u,g}(t)}, \quad (56)$$

where  $\varrho_{J_{u,g}(t)} = \varpi_{u,g}(t)(\mathbf{1}\{J_{u,g}(t)\} + \mathbf{1}\{J_{u,g}^C(t)\})$ ,  $\varpi_{u,g}(t) = \hat{d}_{u,g}(t)\hat{x}_{u,g}^*(t)$ , and  $J_{u,g}^C(t)$  represents the complement of  $J_{u,g}(t)$ . When event  $J_{u,g}^C(t)$  occurs,  $\log_2(1 + \frac{\hat{\Psi}_{u,g}(t)}{\sigma_g^2(t)} \hat{P}_{u,g}^*(t)) \leq \log_2(1 + \frac{\Psi_{u,g}(t)}{\sigma_g^2(t)} \hat{P}_{u,g}^*(t))$  holds. As such, we have

$$\begin{aligned} \Omega_u^2(t) & \leq \sum_{g \in \mathcal{G}_u(t)} \kappa_{\hat{\Psi}_{u,g}(t), \Psi_{u,g}(t)}(\hat{P}_{u,g}^*(t))\varpi_{u,g}(t)\mathbf{1}\{J_{u,g}(t)\} \\ & \stackrel{(a)}{\leq} \frac{B_c \tau}{\ln 2} \sum_{g \in \mathcal{G}_u(t)} \frac{\varpi_{u,g}(t)\hat{P}_{u,g}^*(t)\hat{x}_{u,g}^*(t)\mathbf{1}\{J_{u,g}(t)\}}{\sqrt{(1 + \frac{\hat{\Psi}_{u,g}(t)}{\sigma_g^2(t)} \hat{P}_{u,g}^*(t))(1 + \frac{\Psi_{u,g}(t)}{\sigma_g^2(t)} \hat{P}_{u,g}^*(t))}} \\ & \leq \frac{B_c \tau P_{\max}}{\ln 2} \sum_{g \in \mathcal{G}_u(t)} \frac{\varpi_{u,g}(t)}{\sigma_g^2(t)} \hat{x}_{u,g}^*(t)\mathbf{1}\{J_{u,g}(t)\} \\ & \leq \frac{B_c \tau P_{\max}}{\sigma_{\min}^2 \ln 2} \sum_{g \in \mathcal{G}_u(t)} \varpi_{u,g}(t)\hat{x}_{u,g}^*(t)\mathbf{1}\{J_{u,g}(t)\}, \end{aligned} \quad (57)$$

where  $\iota_{u,g}(t) = \hat{\Psi}_{u,g}(t) - \Psi_{u,g}(t)$ ,  $\sigma_{\min}^2$  is the minimum value of noise power and (a) follows from the Arithmetic-Logarithmic-Geometric Mean Inequality:

$$\frac{a+b}{2} > \frac{b-a}{\ln b - \ln a} > \sqrt{ab}, \text{ if } b > a > 0. \quad (58)$$

Let  $t_{u,g}^i$  denote the time slot of GS  $g$  selected by LEO satellite  $u$  for the  $i$ -th time. Sum (57) over time slots to yield:

$$\begin{aligned} \sum_{t=1}^T \Omega_u^2(t) &\leq \frac{\tau B_c P_{\max}}{\sigma_{\min}^2 \ln 2} \sum_{t=1}^T \sum_{g \in \mathcal{G}_u(t)} \iota_{u,g}(t) \hat{x}_{u,g}^*(t) \mathbf{1}\{J_{u,g}(t)\} \\ &= \frac{\tau B_c P_{\max}}{\sigma_{\min}^2 \ln 2} \sum_{g \in \mathcal{G}_u(t)} \sum_{i=1}^{n_{u,g}(T)} \iota_{u,g}(t_{u,g}^i) \hat{x}_{u,g}^*(t_{u,g}^i) \mathbf{1}\{J_{u,g}(t_{u,g}^i)\} \\ &\leq \frac{\tau B_c P_{\max}}{\sigma_{\min}^2 \ln 2} (G\Psi_0 + \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \phi_{2,u,g}(t_{u,g}^i)), \end{aligned} \quad (59)$$

where  $\Psi_0 = \Psi_{\max} - \Psi_{\min}$  and  $\phi_{2,u,g}(t_{u,g}^i)$  is defined as

$$\phi_{2,u,g}(t_{u,g}^i) = \iota_{u,g}(t_{u,g}^i) \hat{x}_{u,g}^*(t_{u,g}^i) \mathbf{1}\{J_{u,g}(t_{u,g}^i)\}. \quad (60)$$

Furthermore, we define event  $W_{u,g}(t) = \{\hat{\Psi}_{u,g}(t) - \Psi_{u,g}(t) \leq \sqrt{\frac{3 \ln t}{2n_{u,g}(t-1)}}\}$  and combine it with (59) to obtain:

$$\begin{aligned} &\frac{\sigma_{\min}^2 \ln 2}{\tau B_c P_{\max}} \sum_{t=1}^T \Omega_u^2(t) \\ &\leq G\Psi_0 + \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \phi_{2,u,g}(t_{u,g}^i) \mathbf{1}\{W_{u,g}(t_{u,g}^i)\} \\ &\quad + \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \phi_{2,u,g}(t_{u,g}^i) \mathbf{1}\{W_{u,g}^C(t_{u,g}^i)\}. \end{aligned} \quad (61)$$

To obtain the upper bound of the right-hand side of (61), we successively calculate the upper bound of its second term and third term. We further use (21) to obtain the inequality as follows:

$$\begin{aligned} \iota_{u,g}(t) &= \hat{\Psi}_{u,g}(t) - \bar{\Psi}_{u,g}(t) + \bar{\Psi}_{u,g}(t) - \Psi_{u,g}(t) \\ &\leq 2\sqrt{\frac{3 \ln t}{2n_{u,g}(t-1)}}. \end{aligned} \quad (62)$$

Plugging (62) into the second term of the right-hand side of (61), we have

$$\begin{aligned} &\sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \phi_{2,u,g}(t_{u,g}^i) \mathbf{1}\{W_{u,g}(t_{u,g}^i)\} \\ &\leq \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} 2\sqrt{\frac{3 \ln t_{u,g}^i}{2n_{u,g}(t_{u,g}^i-1)}} \\ &\leq \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \sqrt{\frac{6 \ln T}{i}} \stackrel{(b)}{\leq} \sqrt{6GT \ln T}, \end{aligned} \quad (63)$$

where (b) is because

$$\begin{aligned} \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \sqrt{\frac{1}{i}} &\leq \sum_{g \in \mathcal{G}_u(t)} \int_1^{n_{u,g}(T)} \sqrt{\frac{1}{i}} di \\ &\leq 2 \sum_{g \in \mathcal{G}_u(t)} \sqrt{n_{u,g}(T)} \leq 2\sqrt{GT}. \end{aligned} \quad (64)$$

We combine (60) with the definition of  $\Psi_0$  to obtain:

$$\begin{aligned} &\sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \phi_{2,u,g}(t_{u,g}^i) \mathbf{1}\{W_{u,g}^C(t_{u,g}^i)\} \\ &\leq \Psi_0 \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \hat{x}_{u,g}^*(t_{u,g}^i) \mathbf{1}\{W_{u,g}^C(t_{u,g}^i)\}. \end{aligned} \quad (65)$$

By taking expectation of both sides of (65), we have

$$\begin{aligned} &\sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \mathbb{E}\{\phi_{2,u,g}(t_{u,g}^i) \mathbf{1}\{W_{u,g}^C(t_{u,g}^i)\}\} \\ &\leq \Psi_0 \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \mathbb{E}\{\hat{x}_{u,g}^*(t_{u,g}^i)\} \mathbb{E}\{\mathbf{1}\{W_{u,g}^C(t_{u,g}^i)\}\} \\ &\leq \Psi_0 \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \Pr\{W_{u,g}^C(t_{u,g}^i)\}. \end{aligned} \quad (66)$$

According to the Chernoff-Hoeffding bound [73], we obtain

$$\begin{aligned} \Pr\{W_{u,g}^C(t)\} &= \Pr\left\{\hat{\Psi}_{u,g}(t) - \Psi_{u,g}(t) > \sqrt{\frac{3 \ln t}{2n_{u,g}(t-1)}}\right\} \\ &\leq \exp\left(-2n_{u,g}(t-1) \frac{3 \ln t}{2n_{u,g}(t-1)}\right) = t^{-3}. \end{aligned} \quad (67)$$

Plugging (67) into (66), we have

$$\begin{aligned} &\sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \mathbb{E}\{\phi_{2,u,g}(t_{u,g}^i) \mathbf{1}\{W_{u,g}^C(t_{u,g}^i)\}\} \\ &\leq \Psi_0 \sum_{t=2}^T t^{-3} \leq \Psi_0 \int_1^\infty t^{-3} dt \leq \frac{\Psi_0}{2}. \end{aligned} \quad (68)$$

We combine (61), (63) and (68) to yield:

$$\sum_{t=1}^T \Omega_u^2(t) \leq \frac{\tau B_c P_{\max}}{\sigma_{\min}^2 \ln 2} \left( \left(G + \frac{1}{2}\right) \Psi_0 + 2\sqrt{6GT \ln T} \right). \quad (69)$$

c) *Upper bound of  $\Omega_u^3(t)$ :* We recall  $J_{u,g}(t) = \{\Psi_{u,g}(t) < \hat{\Psi}_{u,g}(t)\}$  to obtain

$$\begin{aligned} \Omega_u^3(t) &\leq B_c \sum_{g \in \mathcal{G}_u(t)} \log_2 \left( \frac{1 + \frac{\Psi_{u,g}(t)}{\sigma_{\min}^2(t)} P'_{u,g}(t)}{1 + \frac{\hat{\Psi}_{u,g}(t)}{\sigma_{\min}^2(t)} P'_{u,g}(t)} \right) d'_{u,g}(t) x'_{u,g}(t) \mathbf{1}\{J_{u,g}^C(t)\} \end{aligned}$$

$$\leq -\frac{B_c \tau P_{\max}}{\sigma_{\min}^2 \ln 2} \sum_{g \in \mathcal{G}_u(t)} \iota_{u,g}(t) x'_{u,g}(t) \mathbf{1} \{J_{u,g}^C(t)\}. \quad (70)$$

By taking expectation of the two sides of (70) and summarizing it over time slots, we have

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}\{\Omega_u^3(t)\} &\leq \frac{\tau B_c P_{\max}}{\sigma_{\min}^2 \ln 2} G \Psi_0 + \\ &\frac{\tau B_c P_{\max} \Psi_0}{\sigma_{\min}^2 \ln 2} \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \mathbb{E}\{x'_{u,g}(t_{u,g}^i)\} \mathbf{1} \{J_{u,g}^C(t_{u,g}^i)\} \\ &\leq \frac{\tau B_c P_{\max}}{\sigma_{\min}^2 \ln 2} \Psi_0 \left( G + \sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \Pr \{J_{u,g}^C(t_{u,g}^i)\} \right). \end{aligned} \quad (71)$$

Then, we apply the Chernoff-Hoeffding bound to obtain

$$\begin{aligned} \Pr \{J_{u,g}^C(t)\} &= \Pr \left\{ \Psi_{u,g}(t) - \bar{\Psi}_{u,g}(t) > \sqrt{\frac{3 \ln t}{2n_{u,g}(t-1)}} \right\} \\ &\leq \exp \left( -2n_{u,g}(t-1) \frac{3 \ln t}{2n_{u,g}(t-1)} \right) = t^{-3}. \end{aligned} \quad (72)$$

We further combine (68) with (72) to yield:

$$\sum_{g \in \mathcal{G}_u(t)} \sum_{i=2}^{n_{u,g}(T)} \Pr \{J_{u,g}^C(t_{u,g}^i)\} \leq \frac{1}{2}. \quad (73)$$

Next, we plug (73) into (71) to obtain:

$$\sum_{t=1}^T \mathbb{E}\{\Omega_u^3(t)\} \leq \frac{\tau B_c P_{\max}}{\sigma_{\min}^2 \ln 2} \left( G + \frac{1}{2} \right) \Psi_0. \quad (74)$$

By combining (53), (69) and (74), we have

$$\frac{\sum_{t=1}^T \mathbb{E}\{\Omega_u^1(t)\}}{V \tau B_c P_{\max}} \leq \frac{(2G+1)\Psi_0 + 2\sqrt{6GT \ln T}}{\sigma_{\min}^2 \ln 2}. \quad (75)$$

Finally, we combine (75) with (49) to obtain (26). ■

## APPENDIX C PROOF OF THEOREM 3

Here, we use the solving framework in Appendix A to address **P11**. Note that the partition point of  $d_{u,g}(t)$  is calculated in (9), we have three cases for the range of  $P_0^{u,g}(t)$ :  $P_0^{u,g}(t) \in [P_{\min}, P_{\max}]$ ,  $P_0^{u,g}(t) \in (P_{\max}, \infty)$ , and  $P_0^{u,g}(t) \in (0, P_{\min})$ .

*Case 1:* Since  $P_0^{u,g}(t) \in [P_{\min}, P_{\max}]$ , we can obtain from (9) that the value of  $d_{u,g}(t)$  has two cases:

When  $P_{u,g}(t) \in [P_{\min}, P_0^{u,g}(t)]$ , we have  $d_{u,g}(t) = \tau$  and rewrite **P11** as the following problem:

$$\begin{aligned} \mathbf{P20:} \quad &\min_{P_{u,g}(t)} (Z_u(t)P_{u,g}(t) - Q_u(t)R_{u,g}(t) - VR_{u,g}(t))\tau \\ &\text{s.t. } P_{\min} \leq P_{u,g}(t) \leq P_0^{u,g}(t). \end{aligned}$$

Furthermore, we substitute (1) and (2) into **P20** to yield:  $(Z_u(t)P_{u,g}(t) - (Q_u(t) + V)B_c \log_2(1 + \frac{\Psi_{u,g}(t)}{\sigma_g^2(t)}P_{u,g}(t)))\tau$ .

We further define  $\Phi_{P_{u,g}(t)}^2$  as (31) and deduce its first-order and second-order derivation as follows:

$$\nabla \Phi_{P_{u,g}(t)}^2 = Z_u(t) - \frac{(Q_u(t) + V)B_c \Psi_{u,g}(t)}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{u,g}(t)) \ln 2}, \quad (76)$$

and

$$\nabla^2 \Phi_{P_{u,g}(t)}^2 = \frac{(Q_u(t) + V)B_c (\Psi_{u,g}(t))^2}{(\sigma_g^2(t) + \Psi_{u,g}(t)P_{u,g}(t))^2 \ln 2}. \quad (77)$$

From (77), we see that  $\nabla^2 \Phi_{P_{u,g}(t)}^2 \geq 0$ , indicating that **P20** is convex. Thus, we use  $\nabla \Phi_{P_{u,g}(t)}^2 = 0$  to compute an extremal point to **P20**, given by (29). According to (29), we can obtain an optimal solution to **P20** as (30).

When  $P_{u,g}(t) \in (P_0^{u,g}(t), P_{\max}]$ , we have  $d_{u,g}(t) = \frac{Q_u(t)}{R_{u,g}(t)}$  and rewrite **P11** as the following problem:

$$\begin{aligned} \mathbf{P21:} \quad &\min_{P_{u,g}(t)} \left( Z_u(t) \frac{P_{u,g}(t)}{R_{u,g}(t)} - Q_u(t) - V \right) Q_u(t) \\ &\text{s.t. } P_0^{u,g}(t) \leq P_{u,g}(t) \leq P_{\max}. \end{aligned}$$

We further recast **P21** as the following form:

$$\begin{aligned} \mathbf{P22:} \quad &\min_{\eta_{u,g}(t)} \varphi_{\eta_{u,g}(t)}^2 Q_u(t) \\ &\text{s.t. C9,} \end{aligned}$$

where  $\varphi_{\eta_{u,g}(t)}^2$  is defined as (32). We further derive the first-order and second-order derivation of  $\varphi_{\eta_{u,g}(t)}^2$ , which has the same form of (41) and (42). As such, the optimal power to **P20** is given by (34).

*Case 2:* Since  $P_0(t) \in (P_{\max}, \infty)$ , we have  $d_{u,g}(t) = \tau$  and recast **P11** as the following form:

$$\begin{aligned} \mathbf{P22:} \quad &\min_{P_{u,g}(t)} (Z_u(t)P_{u,g}(t) - Q_u(t)R_{u,g}(t) - VR_{u,g}(t))\tau \\ &\text{s.t. C6.} \end{aligned}$$

It is easy to obtain that  $P_{u,g,2}^0(t)$  is also an extremal point to **P22**, since it has the same form as **P20**. As such, the optimal power to **P22** is given by (35).

Case 3: Since  $P_0(t) \in (0, P_{\min})$ , we have  $d_{u,g}(t) = \frac{Q_u(t)}{R_{u,g}(t)}$  and rewrite **P11** as the following problem:

$$\begin{aligned} \mathbf{P23}: \min_{P_{u,g}(t)} & \left( Z_u(t) \frac{P_{u,g}(t)}{R_{u,g}(t)} - Q_u(t) - V \right) Q_u(t) \\ \text{s.t. } & \text{C6.} \end{aligned}$$

We further transform **P23** into the following form:

$$\begin{aligned} \mathbf{P24}: \min_{\eta_{u,g}(t)} & \varphi_{\eta_{u,g}(t)}^2 Q_u(t) \\ \text{s.t. } & \text{C10.} \end{aligned}$$

Since it has the same form of **P21**, we can obtain that the optimal solution to **P24** is given by

$$P_{u,g}^*(t) = P_{\min}. \quad (78)$$

In brief, we use (34), (35), and (78) to obtain (33). ■

#### APPENDIX D PROOF OF THEOREM 4

Similar to the proof of Theorem 2 in Appendix B, we still use the DPP metric to calculate the drift-plus-regret as follows:

$$\begin{aligned} & \Delta_2(H_u(t)) + V\mathbb{E}\{\Delta\text{Reg}_u(t)\} \\ & \leq B_2 + \mathbb{E}\{Z_u(t)(e_u(t) - e_u^{\text{avg}})|H_u(t)\} + \Omega_u^1(t) \\ & \quad + \mathbb{E}\{Q_u(t)(\lambda(t) - R_u(t)d_u(t))|H_u(t)\}. \end{aligned} \quad (79)$$

Take expectation of the two sides of (79) and summarize it over time slots to yield:

$$\begin{aligned} & \mathbb{E}\{\Delta_2(H_u(T))\} - \mathbb{E}\{\Delta_2(H_u(1))\} + V \sum_{t=1}^T \mathbb{E}\{\Delta\text{Reg}_u(t)\} \\ & \leq TB_2 + \sum_{t=1}^T \mathbb{E}\{\Omega_u^1(t)\}. \end{aligned} \quad (80)$$

We divide the two sides of (80) by  $TV$  to obtain:

$$\mathbb{E}\left\{\frac{1}{T} \sum_{t=1}^T \Delta\text{Reg}_u(t)\right\} \leq \frac{B_2}{V} + \frac{1}{TV} \sum_{t=1}^T \mathbb{E}\{\Omega_u^1(t)\}. \quad (81)$$

We further substitute (25) and (44) into (81) to obtain:

$$\text{Reg}(T) \leq \frac{B_2U}{V} + \frac{1}{TV} \sum_{u \in \mathcal{U}} \sum_{t=1}^T \mathbb{E}\{\Omega_u^1(t)\}. \quad (82)$$

Since (82) has a form similar to (49), we obtain (36). ■

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**Lijun He** (Member, IEEE) received the B.S. degree in electronic information science and technology from Anqing Normal University, Anhui, China, in 2013, and the Ph.D. degree in military communications from the State Key Laboratory of ISN, Xidian University, Xi'an, China, in 2020. From 2018 to 2019, he was with the University of Toronto, Toronto, ON, Canada, as a Visiting Scholar funded by the China Scholarship Council (CSC). From 2020 to 2022, he was a Postdoctoral Researcher with the School of Software, Northwestern Polytechnical University (NPU). From 2022 to 2024, he was an Associate Professor with the School of Software, NPU. He is currently an Associate Professor with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China. His research interests include satellite communication networks, unmanned aerial vehicle networks, and Internet of Things.



**Zheyuan Li** received the B.E. degree in software engineering in 2023 from Northwestern Polytechnical University, Xi'an, China, where he is currently working toward the M.E. degree in software engineering with the School of Software. His research interests include routing, online learning, resource allocation, and non-terrestrial network.



**Juncheng Wang** (Member, IEEE) received the B.Eng. degree in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2014, the M.Sc. degree in electrical and computer engineering from the University of Alberta, Edmonton, AB, Canada, in 2017, and the Ph.D. degree in electrical and computer engineering from the University of Toronto, Toronto, ON, Canada, in 2023. He is currently an Assistant Professor with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China. His research interests include network artificial intelligence, distributed machine learning, and online optimization.



**Ziye Jia** (Member, IEEE) received the B.E., M.S., and Ph.D. degrees in communication and information systems from Xidian University, Xi'an, China, in 2012, 2015, and 2021, respectively. From 2018 to 2020, she was a Visiting Ph.D. Student with the Department of Electrical and Computer Engineering, University of Houston, Houston, TX, USA. She is currently an Associate Professor with the Key Laboratory of Dynamic Cognitive System of Electromagnetic Spectrum Space, Ministry of Industry and Information Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China. Her research interests include space-air-ground networks, aerial access networks, UAV networking, resource optimization, and machine learning.



**Yanting Wang** (Member, IEEE) received the B.S. and Ph.D. degrees in communication and information systems from Xidian University, Xi'an, China, in 2012 and 2019, respectively. She is currently an Assistant Professor with the School of Software, Northwestern Polytechnical University, Xi'an. Her research interests include computation offloading, caching, applications of convex optimization theory, and heterogeneous networks. She was the Technical Program Committee Member for IEEE VTC-2020.



**Chau Yuen** (Fellow, IEEE) received the B.Eng. and Ph.D. degrees from Nanyang Technological University, Singapore, in 2000 and 2004, respectively. He was a Postdoctoral Fellow with Lucent Technologies Bell Labs, Murray Hill, in 2005. From 2006 to 2010, he was with the Institute for Infocomm Research, Singapore. From 2010 to 2023, he was with the Engineering Product Development Pillar, Singapore University of Technology and Design, Singapore. Since 2023, he has been with the School of Electrical and Electronic Engineering, Nanyang Technological University. He is currently the Provost's Chair in wireless communications, and Assistant Dean of Graduate College. He holds 4 US patents and has authored or coauthored more than 400 research papers at international journals. Dr. Yuen received IEEE Communications Society Leonard G. Abraham Prize (2024), IEEE Communications Society Best Tutorial Paper Award (2024), IEEE Communications Society Fred W. Ellersick Prize (2023), IEEE Marconi Prize Paper Award in Wireless Communications (2021), IEEE APB Outstanding Paper Award (2023), and EURASIP Best Paper Award for Journal on Wireless Communications and Networking (2021). He is the Editor-in-Chief of Springer Nature Computer Science, Editor for IEEE Transactions on Vehicular Technology, IEEE System Journal, and IEEE Transactions on Network Science and Engineering, where he was awarded as IEEE TNSE Excellent Editor Award and Top Associate Editor for TVT from 2009 to 2015. He was also the Guest Editor for several special issues, including IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, IEEE WIRELESS COMMUNICATIONS MAGAZINE, IEEE COMMUNICATIONS MAGAZINE, IEEE VEHICULAR TECHNOLOGY MAGAZINE, IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING, and ELSEVIER *APPLIED ENERGY*. He is a Distinguished Lecturer of IEEE Vehicular Technology Society, Top 2% Scientists by Stanford University, and also a Highly Cited Researcher by Clarivate Web of Science.



**Zhu Han** (Fellow, IEEE) received the B.S. degree in electronic engineering from Tsinghua University, Beijing, China, in 1997, and the M.S. and Ph.D. degrees in electrical and computer engineering from the University of Maryland, College Park, MD, USA, in 1999 and 2003, respectively. From 2000 to 2002, he was an R&D Engineer of JDSU, Germantown, MD, USA. From 2003 to 2006, he was a Research Associate with the University of Maryland. From 2006 to 2008, he was an Assistant Professor with Boise State University, Boise, ID, USA. He is currently a John and Rebecca Moores Professor with the Electrical and Computer Engineering Department as well as with the Computer Science Department, the University of Houston, Houston, TX, USA. His main research focuses on the novel game-theory related concepts critical to enabling efficient and distributive use of wireless networks with limited resources. His other research interests include wireless resource allocation and management, wireless communications and networking, quantum computing, data science, smart grid, carbon neutralization, security, and privacy. Dr. Han received an NSF Career Award in 2010, Fred W. Ellersick Prize of IEEE Communication Society in 2011, EURASIP Best Paper Award for the Journal on Advances in Signal Processing in 2015, IEEE Leonard G. Abraham Prize in the field of Communications Systems (best paper award in IEEE JSAC) in 2016, IEEE Vehicular Technology Society 2022 Best Land Transportation Paper Award, and several best paper awards in IEEE conferences. He was an IEEE Communications Society Distinguished Lecturer from 2015 to 2018 and ACM Distinguished Speaker from 2022 to 2025, AAAS fellow since 2019, and ACM fellow since 2024. He is a 1% highly cited Researcher since 2017 according to Web of Science. He is also the winner of the 2021 IEEE Kiyo Tomiyasu Award (an IEEE Field Award), for outstanding early to mid-career contributions to technologies holding the promise of innovative applications, with the following citation: "for contributions to game theory and distributed management of autonomous communication networks."